

Unit-1 : Set Theory .

Q1. Define sets, subsets and proper subsets with appropriate examples.

→ A set is a well-defined collection of distinct objects or elements.

for eg:

$A = \{2, 4, 6, 8\}$ or {the set of even number less than 10}

$B = \{\text{red, blue, green}\}$

A set A is called subset of another set B if every element of A is also an element of B . i.e. $A \subseteq B$.

for eg:

$A = \{1, 2\}$

$B = \{1, 2, 3, 4\}$

since every element of A is in B , $A \subseteq B$.

A proper subset is a subset that is not equal to the original set. It contains some, but not all, elements of the larger set i.e. $A \subset B$.

for eg:

$A = \{2, 4\}$

$B = \{2, 4, 6, 8\}$

Here, every element of A is in B , and $A \neq B$, Therefore, A is a proper subset of B .

Q2. Differentiate between a universal set, an empty set and disjoint sets.

→ Universal set is a big set that contains all the objects or elements where it should be well-defined.

Whereas, Empty set is the set that has no objects or elements inside it.

While, Disjoint sets is the two or more sets that do not share any of the same items or elements.

Likewise,

Universal set is represented by U .

Empty set is represented by ϕ or $\{ \}$.

Disjoint set is represented by $A \cap B = \phi$.

Q3. Explain the Set operations: Union, intersection and difference with venn-diagrams.

→ The set operations :-

1. Union ($\cup$)

→ The union of two sets contains all elements that are in either sets or in both sets.
Symbol: $A \cup B$.

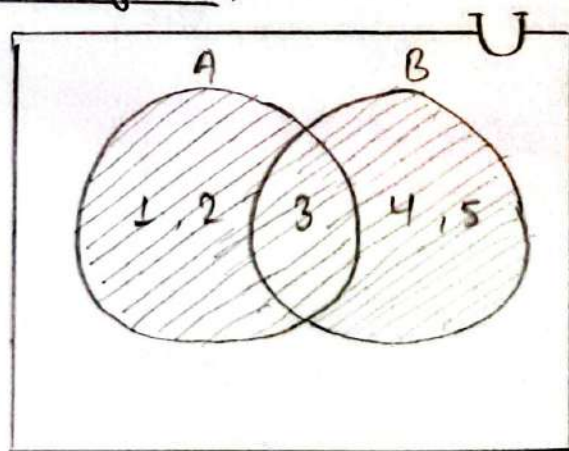
For eg:

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$\therefore A \cup B = \{1, 2, 3, 4, 5\}.$$

Venn-diagram :-



2) Intersection ($\cap$)

→ The intersection set is the overlapping set of the two sets that contains only the elements common to both sets.

Symbol: $A \cap B$.

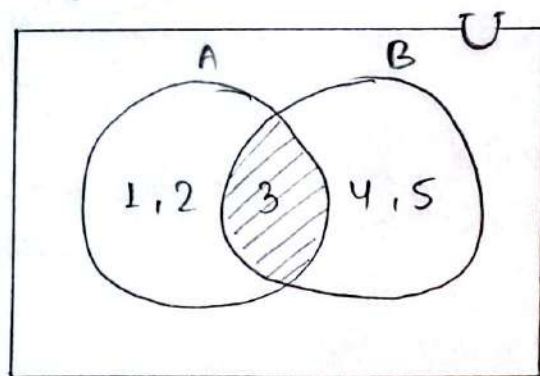
For eg:

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$\therefore A \cap B = \{3\}$$

Venn-diagram:



3) Difference ($-$)

→ The difference of two sets contains elements that are in the first set but not in the second set.

Symbol: $A - B$.

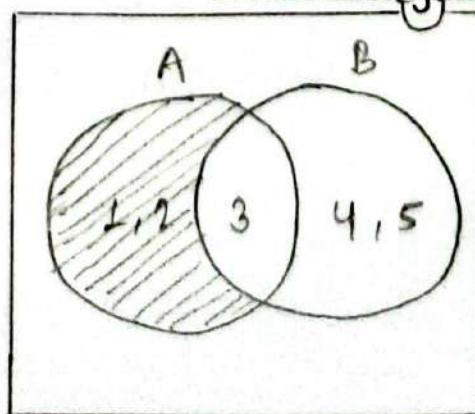
For eg:

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$\therefore A - B = \{1, 2\}$$

Venn-diagram:



Q4. What is the symmetric difference of two sets? Find $A \oplus B$ if $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$.

→

The symmetric difference of two sets is the set of elements that belong to either of the sets but not to both.

Symbol: $A \oplus B$.

It can be written as:

$$A \oplus B = (A - B) \cup (B - A)$$

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ then,

$$\begin{aligned} A \oplus B &= (A - B) \cup (B - A) \\ &= \{1, 2\} \cup \{5, 6\} \end{aligned}$$

$$\therefore A \oplus B = \{1, 2, 5, 6\}.$$

Q5. State and prove De Morgan's laws for sets using algebraic set identities.

→

The De-morgan's 1st law states that the complement of union of two sets is equivalent to the intersection of complement of individual sets.

$$\text{i.e. } \overline{A \cup B} = \bar{A} \cap \bar{B}.$$

Let $p = \overline{A \cup B}$ and $q = \bar{A} \cap \bar{B}$.

Let x be an arbitrary elements of p then,

$$\begin{aligned} x &\in p. \\ &= x \in \overline{A \cup B} \\ &= x \notin A \cup B \\ &= x \notin A \text{ OR } x \notin B \\ &= x \in \bar{A} \text{ and } x \in \bar{B} \\ &= x \in \bar{A} \text{ and } \bar{B} \\ &= x \in \bar{A} \cap \bar{B} \\ &= x \in q \end{aligned}$$

Therefore, $p \subset q$ ——— (i)

Again,

Let y be an arbitrary element of Q then,

$$\begin{aligned} & y \in Q \\ &= y \in \bar{A} \cap \bar{B} \\ &= y \in \bar{A} \text{ and } \bar{B} \\ &= y \notin A \text{ OR } y \notin B \\ &= y \notin A \cup B \\ &= y \notin \overline{A \cup B} \\ &= y \in P \end{aligned}$$

Therefore, $Q \subset P$ — (ii)

Now, combining eqn (i) and (ii), we get, $P = Q$.

$$\text{P.e. } \overline{A \cup B} = \bar{A} \cap \bar{B}.$$

Demorgan's 2nd law states that the complement of intersection of two sets is equivalent to the union of complement of individual sets.

$$\text{P.e. } \overline{A \cap B} = \bar{A} \cup \bar{B}.$$

$$\text{Let } M = \overline{A \cap B} \text{ and } N = \bar{A} \cup \bar{B}.$$

Let x be an arbitrary elements of M then,

$$\begin{aligned} & x \in M \\ &= x \in \overline{A \cap B} \\ &= x \notin A \cap B \\ &= x \notin A \text{ and } B \\ &= x \notin A \text{ and } x \notin B \\ &= x \notin \bar{A} \text{ OR } x \notin \bar{B} \\ &= x \in \bar{\bar{A}} \text{ OR } x \in \bar{\bar{B}} \\ &= x \in \bar{A} \cup \bar{B} \\ &= x \in N \end{aligned}$$

Therefore, $M \subset N$ — (i)

Again,

Let y be an arbitrary element of N then,
 $y \in N$.

$$= y \in \bar{A} \cup \bar{B}$$

$$= y \in \bar{A} \text{ OR } \bar{B}$$

$$= y \in \bar{A} \text{ OR } y \in \bar{B}$$

$$= y \notin A \text{ and } y \notin B$$

$$= y \notin A \text{ and } B$$

$$= y \in \overline{A \cap B}$$

$$= y \in M$$

Therefore, $N \subset M$ — (ii)

Now, combining (i) and (ii) we get, $M = N$.

∴ $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Q6. Define the cartesian product of two sets. And
 $A \times B$ if $A = \{a, b\}$ and $B = \{1, 2, 3\}$

→ The cartesian product of two sets is the set of all ordered pairs where the first element of set A and first element of set B make pair and so on.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

If $A = \{a, b\}$ and $B = \{1, 2, 3\}$

then,

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}.$$

Q7) Define a power set. Compute the power set of $S = \{x, y, z\}$.

→ The power set of a set is the set of all possible subsets of that set, including the empty set and the set itself.

$$\text{i.e. } P(A) = \{x \mid x \subseteq A\}.$$

If $S = \{x, y, z\}$ then,

$$P(S) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{x, z\}, \{x, y, z\}\}.$$

Q8) IF a set has n elements, prove that its power set has 2^n elements.

→ proof:

Let a set A have n elements.

The power set of A ($P(A)$) is the set of all subsets of A .

For each element of A , there are two choices when forming a subset:

1. If include element.
2. If do not include the element.

Since, there are n elements and each has 2 independent choices, the total number of possible subsets is:

$$2 \times 2 \times 2 \dots \times 2 \text{ (n times)} = 2^n.$$

Hence, $|P(A)| = 2^n$.

Q9) prove the distributive law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

→ Let x be any arbitrary element.

Taking L.H.S,

Suppose,

$$\begin{aligned} x &\in A \cup (B \cap C) \\ &= x \in A \text{ OR } (B \cap C) \\ &= x \in A \text{ OR } x \in (B \cap C) \end{aligned}$$

$$\begin{aligned}
 &= x \in A \text{ OR } x \in B \text{ and } x \in C \\
 &= x \in A \text{ OR } x \in B \text{ and } x \in A \text{ OR } x \in C \\
 &= x \in A \text{ OR } B \text{ and } x \in A \text{ OR } C \\
 &= x \in (A \cup B) \text{ and } x \in (A \cup C) \\
 &= x \in (A \cup B) \cap (A \cup C)
 \end{aligned}$$

Thus, $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ — (i)

Again,

Taking RHS,

Let $x \in (A \cup B) \cap (A \cup C)$.

$$\begin{aligned}
 &= x \in (A \cup B) \text{ and } x \in (A \cup C) \\
 &= x \in A \text{ OR } B \text{ and } x \in A \text{ OR } C \\
 &= x \in A \text{ OR } x \in B \text{ and } C \\
 &= x \in A \text{ OR } x \in B \cap C \\
 &= x \in A \cup (B \cap C)
 \end{aligned}$$

Thus, $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$ — (ii)

Now, combining (i) and (ii), we get,

$$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Q10) prove the absorption law: $A \cup (A \cap B) = A$.

→ Let x be any arbitrary element.

Taking LHS.

Let $x \in A \cup (A \cap B)$ then,

$$x \in A \text{ OR } x \in A \cap B$$

If $x \in A \cap B$ then,

$$x \in A$$

Hence, $x \in A$.

Therefore, $A \cup (A \cap B) \subseteq A$.

Now,

if $x \in A$, then obviously,
 $x \in A \cup (A \cap B)$.

So,

$$A \subseteq A \cup (A \cap B)$$

Since, both sets are subsets of each other,
we can say that,

$$A \cup (A \cap B) = A$$

Hence, proved.

Q11. Show that $A - B = A \cap \bar{B}$.

→ proof:

By definition,

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

The complement of B is

$$\bar{B} = \{x : x \notin B\}.$$

Now,

$$\begin{aligned} A \cap \bar{B} &= \{x : x \in A \text{ and } x \in \bar{B}\} \\ &= \{x : x \in A \text{ and } x \notin B\}. \end{aligned}$$

This is exactly the definition of $A - B$.

Thus, $A - B = A \cap \bar{B}$.

Hence, proved.

Q12. Construct a Venn diagram for the expression $(A \cup B) \cap \bar{C}$.

→ The shaded part represents all elements that are in A or B but not in C.

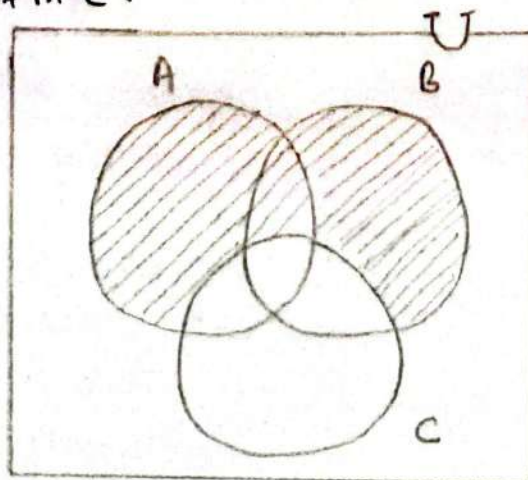


Fig: Venn-diagram of $(A \cup B) \cap \bar{C}$.

Q13. What is the cardinality of a set? find the cardinality of $A \cup B$ if $|A| = 15$, $|B| = 20$, and $|A \cap B| = 5$.

→ The cardinality of a set is the number of elements contained in that set.

Given,

$$|A| = 15$$

$$|B| = 20$$

$$|A \cap B| = 5$$

Now,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 15 + 20 - 5$$

$$= 30$$

$$\therefore |A \cup B| = 30.$$

Q14. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$. Find $\overline{A \cup B}$.

→

Given,

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$A = \{1, 2, 3, 4\}$$

$$B = \{2, 4, 6, 8\}$$

Now,

$$A \cup B = \{1, 2, 3, 4\} \cup \{2, 4, 6, 8\}$$

$$= \{1, 2, 3, 4, 6, 8\}$$

Then,

$$\overline{A \cup B} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} - \{1, 2, 3, 4, 6, 8\}$$

$$= \{5, 7, 9\}$$

$$\therefore \overline{A \cup B} = \{5, 7, 9\}.$$

Q15. Prove that if $A \subseteq B$, then $A \cup B = B$ and $A \cap B = A$.

→ Given,

$$A \subseteq B$$

This means every element of A is also element of B .

Now,

Let $x \in A \cup B$, where x is an arbitrary element.

Then,

either $x \in A$ or $x \in B$.

Since, $A \subseteq B$ if $x \in A$ then $x \in B$.

Hence, every element of $A \cup B$ belongs to B .

So, $A \cup B \subseteq B$

Also,

Every element of B obviously belongs to $A \cup B$.

Thus, $B \subseteq A \cup B$.

Hence, both are subset of each other.

Therefore, $A \cup B = B$.

Again,

Let y be an arbitrary element that belongs to

$A \cap B$.

i.e. $y \in A \cap B$.

Then,

$y \in A$ and $y \in B$.

Thus, $A \cap B \subseteq A$.

Now,

Let $y \in A$.

Since $A \subseteq B$, $y \in B$.

Therefore,

$y \in A \cap B$.

Hence, $A \subseteq A \cap B$.

Here, both are subset of each other.
Therefore,

$A \cap B = A$.

Hence, proved.

Q16. What are partitions of a set? provide an example of partitioning the set of integers.

A partition of a set is a collection of non-empty disjoint subsets whose union equals the original set.

For eg:-

Let 'Z' be the set of integers.

$$P_Z. Z = \{-3, -2, -1, 0, 1, 2, 3, \dots\}.$$

partition of Z into:-

Even integers

$$E = \{-4, -2, 0, 2, 4, \dots\}$$

and

odd integers

$$O = \{-3, -1, 1, 3, 5, \dots\}$$

Here,

$$E \cup O = \phi$$

$$E \cup O = Z$$

Therefore, $\{E, O\}$ is a partition of the set of integers.

Q17. Use set builder notation to describe the set of all odd positive integers less than 20.

The odd positive integers less than 20 are.

$$\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}.$$

Now,

set-builder notation:-

$$\{x \in \mathbb{N} : x < 20, x \text{ is odd number}\}$$

OR,

$$\{2n-1 : n \in \mathbb{N}, 1 \leq n \leq 10\}.$$

Q18.

→

Discuss the concept of bit strings in representing sets and compute the bitwise OR and AND for A and B.

A bit string is a sequence of binary digits (0 or 1) used to represent a set where '1' means element is present in the set and '0' means element is absent from the set.

Representation of sets using bit strings:
Suppose the universal set is

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$\text{let, } A = \{1, 3, 5, 7\}$$

$$B = \{2, 3, 6, 7\}$$

Bit-string representation are:

Element	1	2	3	4	5	6	7	8
A	1	0	1	0	1	0	1	0
B	0	1	1	0	0	1	1	0

Hence,

$$A = \{10101010\} \text{ and } B = \{01101100\}$$

Bitwise OR (A OR B): It gives 1 if either bit is 1.

A	1	0	1	0	1	0	1	0
B	0	1	1	0	0	1	1	0
OR	1	1	1	0	1	1	1	0

Therefore, $A \text{ OR } B = 11101110$ and corresponding set is

$$A \text{ OR } B = \{1, 2, 3, 5, 6, 7\}.$$

Bitwise AND (A AND B): It gives 1 if both bit is 1.

A	1	0	1	0	1	0	1	0
B	0	1	1	0	0	1	1	0
AND	0	0	1	0	0	0	1	0

Therefore, $A \text{ AND } B = 00100010$ and corresponding set is

$$A \text{ AND } B = \{3, 7\}.$$

Q19 Explain the application of sets in database relational models.

→ In the relational database model, a relation (table) is viewed as a set of tuples (rows).

Application of sets in database relational models:

1. Representation of tables.

→ A table is represented as a set of rows where each row is unique, duplicate tuples are not allowed.

Foreg:

Roll	Name
1	Dipen
2	Abhilesh

2. Used in SQL.

→ Sets are used in SQL joins, selection and projection.

If combines every tuple of one table with every tuple of another table. selection retrieves rows satisfying a condition. projection retrieves selection columns.

3. Set operations.

→ Set theory operations are widely used in database query language to combine and manipulate data from one or more tables. It includes:

1. Union ($\cup$): It combines the result of two queries into single result set while eliminating duplicate.
2. Intersection ($\cap$): It retrieves only the record that are common to both queries.
3. Difference ($-$): It returns records that are present in one query but not in another.

Q20. Prove that $(A-B)-C = A-(B \cup C)$

→ proof:-

using the identity,

$$A-B = A \cap \bar{B}.$$

Then, taking L.H.S,

$$\begin{aligned}
 &= (A-B)-C \\
 &= (A \cap \bar{B}) \cap \bar{C} \\
 &= A \cap (\bar{B} \cap \bar{C}) \quad [\text{using associative law}] \\
 &= A \cap (\overline{B \cap C}) \quad [\text{using De Morgan's law}] \\
 &= A - (B \cap C) \quad [\text{using } (A-B) = A \cap \bar{B} \text{ rule}]
 \end{aligned}$$

Hence, $(A-B)-C = A - (B \cap C)$. proved.

Q2.

→ Show that $(A \cap B) \subseteq A \subseteq (A \cup B)$.
Proof:

At first we need to show that $(A \cap B) \subseteq A$.

Let $x \in A \cap B$, is an arbitrary element of $(A \cap B)$.

then, $x \in A$ and $x \in B$.

Therefore, $A \cap B \subseteq A$

Now,

we need to show that $A \subseteq (A \cup B)$.

Let x is an arbitrary element of A .

i.e. $x \in A$

Since, the union contains all elements of either set
 $x \in A \cup B$.

Hence, $A \subseteq (A \cup B)$

Since both statement are true, we can say that,

$$(A \cap B) \subseteq A \subseteq (A \cup B)$$

Hence proved.

Unit-5: Combinatorics and Counting Principles.

Unit-2: Logic and propositional calculus.

Q22 Define proposition, simple proposition and compound proposition with examples.

→ A proposition is a declarative statement that is either true or false but not both.

For eg:

p: Kathmandu is the capital city of Nepal $\rightarrow$ True.

Simple proposition is a proposition that cannot be divided into two or more propositions using logical connectives.

For eg:

p: "Dipen is a student".

Compound proposition is a proposition which is formed by combining two or more simple propositions using logical connectives such as AND, OR, NOT etc.

For eg:

p: "Dipen is a student."

q: "Dipen is programmer".

Then

$p \wedge q$: "Dipen is a student and programmer".

Q23 List the fundamental logical connectives with their truth tables (AND, OR, NOT, XOR).

→

The fundamental connectives are:

1. Conjunction (AND)

→ The AND operation is true only when both p and q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Fig: Truth table of AND.

2) Disjunction (OR)

→ OR operation is false only when both p and q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Fig: Truth table of OR

3) Negation (NOT)

→ The NOT operation reverses the truth value of a proposition.

p	$\neg p$
T	F
F	T

Fig: Truth table of NOT.

4) Exclusive-OR (XOR)

→ The XOR operation is true when one of the operand p and q is true.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Fig: Truth table of XOR.

Q24)

Define implication (conditional) and biconditional statement with truth table.

→ An implication is a compound statement which is false only when the antecedent (p) is true and the consequent (q) is false. Otherwise, it is true.
It is written as $p \rightarrow q$ ("if p then q").

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Fig: Truth table of implication.

Biconditional statements is a compound statement formed by connecting two statements P and q with if and only if.

It is true only when both statements have same value (both true or both false).

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Fig: Truth table of Biconditional.

Q25. What are tautologies, contradictions and contingencies? Give one example of each.

→ A compound proposition that is always true for all possible truth values of its propositional variables is called Tautology.

For eg:
"It is raining or it is not raining" → $P \vee \neg P$.
Truth table.

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

A compound proposition that is always false for all possible truth values of its propositional variables. is called contradiction.

For eg: "It is raining and it is not raining" $\rightarrow p \wedge \neg p$.

Truth table:

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

A compound proposition that is neither tautology nor contradiction, meaning its outcome contains both true and false values. is called contingency.

For eg: "If it is raining, then the ground is wet."

$p \rightarrow q$

Truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Q26.

Show that $p \rightarrow q$ is logically equivalent to $\neg p \vee q$ using truth table.

$\rightarrow$ To prove $p \rightarrow q \equiv \neg p \vee q$.

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Hence, both truth values of $p \rightarrow q$ and $\neg p \vee q$ are equal. So, $p \rightarrow q$ is logically equivalent to $\neg p \vee q$.

Get state and prove the associative and distributive laws of logic.

→ Associative laws state that the grouping of propositions does not affect the truth value when applying the same operation (all conjunctions or all disjunctions).

Associative law of conjunction (A):

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

Associative law of disjunction (V):
 $(p \vee q) \vee r \equiv p \vee (q \vee r)$

These can be proved by truth tables.

P	Q	R	$p \wedge q$	$q \wedge r$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Fig:- Truth table for Associative law of conjunction.

P	Q	R	$p \vee q$	$q \vee r$	$(p \vee q) \vee r$	$p \vee (q \vee r)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	T	T	T
F	F	F	F	F	F	F

Fig:- Truth table for Associative law of Disjunction.

Distributive laws states that how a logical operation distributes over a different logical operation.

Distributive law of conjunction over disjunction:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

Distributive law of Disjunction over conjunction:-

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

proof through truth table.

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	T	T	F
F	T	F	T	F	T	F	F
F	F	T	T	F	F	T	F
F	F	F	F	F	F	F	F

Fig: Truth table of Distributive law of conjunction.

P	Q	R	$Q \vee R$	$P \vee (Q \wedge R)$	$P \vee Q$	$P \vee R$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	F	F	F	F	F

Fig: Truth table of Distributive law of Disjunction.

Q28.

state De-Morgan's laws for propositional logic and verify them using truth tables.

De-morgan's first law states that the negation of a conjunction is the disjunction of the negations.
 $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

De-morgan's second law states that the negation of a disjunction is the conjunction of the negations.
 $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

1. Verification of 1st law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

P	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

2. Verification of 2nd law: $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

P	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Hence, both De-morgan's laws are verified because the corresponding columns in each truth table are identical.

Q29.

Find the converse, inverse and contrapositive of the statement: "If it rains, the ground gets wet".

→ Let,

P: If it rains.

Q: The ground gets wet.

The given statement is:

$P \rightarrow Q$.

∴ Converse.

→ It is obtained by interchanging p and q .
i.e. $q \rightarrow p$.

Statement: If the ground gets wet, then it rains.

2. Inverse.

→ It is obtained by negating both p and q .
i.e. $\neg p \rightarrow \neg q$.

Statement: If it does not rain, then the ground does not get wet.

3. Contrapositive.

→ It is obtained by interchanging and negating both propositions.
i.e. $\neg q \rightarrow \neg p$

Statement: If the ground does not get wet, then it does not rain.

Q30. prove that an implication is logically equivalent to its contrapositive.

→ Let, p and q are two propositions then,

The implication be: $p \rightarrow q$

and,

its contrapositive is: $\neg q \rightarrow \neg p$.

Now, taking implication first,

$$p \rightarrow q \equiv \neg p \vee q \quad [\text{conditional identity}]$$

$$\neg p \vee q \equiv q \vee \neg p \quad [\text{commutative law}]$$

$$q \vee \neg p = \neg(\neg q) \vee \neg p \quad [\text{Double negation law}]$$

$$\neg(\neg q) \vee \neg p \equiv \neg q \rightarrow \neg p \quad [\text{conditional identity}]$$

i.e. $A \rightarrow B \equiv \neg A \vee B$

Therefore, $p \rightarrow q \equiv \neg q \rightarrow \neg p$
proved.

Q31.

Translate the following into propositional logic.

"You can access the Internet from campus only if you are a computer science major or you are not freshman".

→

Let,

p: You can access the Internet from campus.

q: You are a computer science major.

r: You are a freshman.

Given,

$$p \rightarrow (q \vee \neg r).$$

∴ Therefore, propositional logic is $p \rightarrow (q \vee \neg r)$.

Q32.

Define Universal and existential Quantifiers. Provide examples using predicate.

→

Universal quantifiers states that a predicate is true for every element in the domain.

For eg:

Let $P(x)$ be: x belongs to real number.

$$\forall x \in \mathbb{R}, x^2 \geq 0.$$

For every real number x , x^2 is greater than or equal to zero.

Existential quantifiers states that there is at least one element in the domain for which the predicate is true.

For eg:

Let $P(x)$ be: x be an even number.

then,

$$\exists x \in \mathbb{Z}, P(x)$$

There exists an integer x that is even.

Q33. Negate the quantified statement: $\forall x \exists y (p(x,y) \rightarrow q(x))$

→ Given:

$$\forall x \exists y (p(x,y) \rightarrow q(x)).$$

We can use the rules:

$$\neg (\forall x p(x)) \equiv \exists x \neg (p(x))$$

$$\neg (\exists y p(y)) \equiv \forall y \neg p(y).$$

Now,

We can say that,

$$\neg [\forall x \exists y (p(x,y) \rightarrow q(x))] \equiv \exists x \forall y \neg (p(x,y) \rightarrow q(x))$$

$$\text{Since, } \neg(p \rightarrow q) \equiv p \wedge \neg q$$

We get,

$$\exists x \forall y [p(x,y) \wedge \neg q(x)]$$

$$\text{Therefore, } \forall x \exists y (p(x,y) \rightarrow q(x)) \equiv \exists x \forall y [p(x,y) \wedge \neg q(x)].$$

Q34) Explain Modus Ponens and Modus Tollens rules of Inference with examples.

→ Modus Ponens is a valid rule of inference in which we have a conditional statement and confirm its antecedent to conclude its consequent.

$$\text{P.e. } p \rightarrow q$$

$$\frac{p}{\therefore q}$$

For eg,

If I study, then I will pass.

P: I study.

Therefore, I will pass.

Symbol:

$$p \rightarrow q$$

$$p$$

$$\frac{p}{\therefore q}.$$

Modus Tollens is a valid rule of inference in which we have a conditional statement and deny its consequent to conclude that its antecedent is also false.

Foregi

$p \rightarrow q$: If it rains, then the ground gets wet.

$\neg q$: The ground is not wet.

Therefore, $\neg p$: It did not rain.

Q35. Explain the hypothetical syllogism and disjunctive syllogism rules of inference.

→ Hypothetical syllogism is a rule of inference where a conditional conclusion is drawn from two conditional premises.

If statement p leads to q and statement q leads to r , then it can be concluded that the statement p leads to statement r .

$$\begin{array}{l} \text{P} \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Disjunctive syllogism is a rule of inference where a conclusion is reached by eliminating one of the possibilities in an either/or statement.

If you are given that either statement p or statement q must be true and if one of them is false, then the remaining statement must be true.

$$\begin{array}{l} \text{P} \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

Q36.

Determine the validity of the argument: If I study, I will pass. I passed. Therefore I studied.

→

Given:

Let,

p : I study.

q : I will pass.

Then,

Premise 1: $p \rightarrow q$ (If I study, I will pass).

Premise 2: q (I passed).

Therefore: $\therefore p$ (I studied).

This argument is called affirming the consequent, which is fallacy.

Hence, The argument is invalid.

Q37.

Explain the direct proof method with a mathematical example. (e.g., sum of two even no. is even).

→

A direct proof is a method of proving a statement $p \rightarrow q$ by assuming that p is true and logically deriving q .

Example: The sum of two even integers is even. Proof:

Let a and b be two even integers.

by the definition of an even integers.

$a = 2m$ and $b = 2n$ where $m, n \in \mathbb{Z}$.

Now,

$$a + b = 2m + 2n$$

$$= 2(m + n)$$

$$= 2k \quad (\because m + n = k \in \mathbb{Z})$$

Therefore, $a + b$ is an even as $a + b = 2k$.

Hence, The sum of two even integers is even.

→

Explain the method of proof by contradiction. Prove that $\sqrt{2}$ is irrational.

Proof by contradiction is a method in which we assume that the statement to be proved is false and then show that this assumption leads to contradiction.

Given, $\sqrt{2}$ is irrational.

Proof:-

Let us assume that $\sqrt{2}$ is rational number, then it can be expressed as:

$$\sqrt{2} = \frac{p}{q}$$

where, $q \neq 0$ and $p, q \in \mathbb{Z}$ having no common factor.

Squaring both sides, we get.

$$(\sqrt{2})^2 = \left(\frac{p}{q}\right)^2$$

$$\text{or, } 2 = \frac{p^2}{q^2}$$

$$\therefore p^2 = 2q^2 \quad \text{--- (1)}$$

Here, p^2 is even so p must be even.

Let, $p = 2k$.

Substituting p in eqn (1).

$$(2k)^2 = 2q^2$$

$$\text{or, } 4k^2 = 2q^2$$

$$\therefore q^2 = 2k^2$$

Again, q^2 is also even, so q must be even.

This means that p and q have common factor 2 as both are even which contradicts our assumption that p/q is rational. Therefore $\sqrt{2}$ is irrational number.

Q39. What is a proof by contrapositive? prove that if $3n+2$ is odd then n is odd.

→ Proof by contrapositive is a method used to prove an implication by proving its contrapositive.
i.e. $p \rightarrow q \equiv \neg q \rightarrow \neg p$

Given, If $3n+2$ is odd then n is odd.

proof

Let us assume n is even.
then, by definition,

$$n = 2k \quad (k \in \mathbb{Z})$$

Now,

$$\begin{aligned} 3n+2 &= 3(2k)+2 \\ &= 6k+2 \\ &= 2(3k+1) \\ &= 2m \quad (m = (3k+1) \in \mathbb{Z}) \end{aligned}$$

$\therefore 3n+2 = 2m$ is even.

Therefore, n is even $\rightarrow 3n+2$ is also even. i.e. $7q \rightarrow 7p$.

Hence, $7q \rightarrow 7p \equiv p \rightarrow q$ i.e. If $3n+2$ is odd then n is odd.

Q40) Describe proof by cases (exhaustive proof) and provide short example.

→ Proof by cases is a method in which a problem is divided into different possible cases and the statement is proved for each case.

For eg:- Every integer is $n, n+1$ is even.
Every integer is either even or odd.

Case 1: n is even

Let, $n = 2k$ ($k \in \mathbb{Z}$)
Then,

$$\begin{aligned} n(n+1) &= 2k(2k+1) \\ &= 2(2k^2+k) \\ &= 2m \quad [m = 2k^2+k \text{ is also integer}] \end{aligned}$$

$\therefore n(n+1)$ is a even.

Case 2: n is odd.

Let,

$$n = 2k+1 \quad (k \in \mathbb{Z})$$

Then,

$$\begin{aligned} n(n+1) &= (2k+1)(2k+1+1) \\ &= (2k+1)(2k+2) \\ &= 2[(k+1)(2k+1)] \end{aligned}$$

$\therefore n(n+1)$ is also even if n is odd as it has
Common factor 2.

Since both possible cases give an even result.

$\therefore n(n+1)$ is even for every integer n .

Q41) Check if the compound proposition $(p \vee q) \wedge (\neg p \vee r)$
 $\rightarrow (q \vee r)$ is tautology.

$\rightarrow$

Given, $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$

Let, $A = (p \vee q) \wedge (\neg p \vee r)$

$B = (q \vee r)$

then, $A \rightarrow B$.

Case 1: p is true

If p is true then $p \vee q$ is also true.

and

$$\neg p \vee r = \text{false} \vee r = r$$

Since, A need to be true, so r must be true.

Hence, $q \vee r$ is also true

Therefore, $A \rightarrow B$ is true.

Moreover,

Case 2: p is false.

If p is false, then

$\neg p$ is true.

So, $\neg p \vee r$ is also true

Now,

for A need to be true, q must be true. So,

$$\therefore p \vee q = \text{false} \vee q = q \text{ is true.}$$

Then,

$q \vee r$ is also true $\therefore q$ is true

Therefore, $A \rightarrow B$ is true.

Since, the proposition is true for all possible cases.

Thus, $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.

Q42. Differentiate between propositional logic and predicate logic.

→

Propositional logic

predicate logic

1. It deals with propositions that are entirely true or false.

1. It deals with predicates objects and variables.

2. It does not use quantifiers like $\forall x$ or $\exists x$.

2. It uses universal ($\forall x$) and existential ($\exists x$) quantifiers.

3. For eg: $P \wedge Q$, $A \vee B$ etc.

3. For eg: $\forall x (p(x) \rightarrow q(x))$, $\exists x (p(x) \wedge q(x))$ etc.

Unit 3: Relations and Functions.

Q43.

Define a Binary relation. Explain domain, co-domain and range of a relation.

→

A binary relation R from a set A to set B is a subset of the cartesian product $A \times B$.

i.e. $R \subseteq A \times B$.

for eg:

$A = \{1, 2\}$, $B = \{2, 3\}$

$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$

$\therefore R$ is a binary relation from A to B .

Domain is the set of all first elements of the ordered pairs in R .

Domain $(R) = \{1, 2\}$.

co-domain is the entire set of B to which the relation maps or make ordered pair with first elements (domain).

co-domain $(R) = \{2, 3\}$.

Range is the set of all second elements that actually occur in the relation.

Range $(R) = \{2, 3\}$.

$\therefore$ Range $\subseteq$ co-domain.

Q44.

Explain reflexive, symmetric and transitive properties of relation with examples.

→

Reflexive relation R is said to be relation on set A if every element is related to itself.

i.e. $\forall a \in A, (a, a) \in R$.

Symmetric relation is said to be relation R on set A if whenever a is related to b then b is also related to a .

i.e. $(a, b) \in R \rightarrow (b, a) \in R$.

Transitive relation is said to be relation R on set A if a element ' a ' is related to ' b ' and ' b ' is related to ' c ' then, ' a ' is also related to ' c '.

Ex.

$$(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$$

For eg;

$$\text{let } A = \{1, 2, 3\}$$

$$R = \{(1,1), (2,2), (3,3), (4,2), (2,1), (2,3), (4,3)\}$$

Here,

$$(1,1), (2,2), (3,3) \in R$$

$\therefore$ It is ~~then~~ reflexive relation.

And,

$$(1,2), (2,1) \in R$$

$\therefore$ It is symmetric relation.

Similarly,

$$(1,2), (2,3), (1,3) \in R$$

$\therefore$ It is transitive relation.

Q45. What is an anti-symmetric relation? How does it differ from an asymmetric relation?

A relation R on set A is anti-symmetric if whenever both aRb and bRa are true then $a=b$.

$$\text{i.e. } (a,b) \in R \wedge (b,a) \in R \rightarrow (a=b)$$

For eg:-

The relation $\leq$ on Integer is anti-symmetric

If, $a \leq b$ and $b \leq a$ then, $a=b$

Therefore, $\leq$ is anti-symmetric.

A relation R on set A is asymmetric if $(a,b) \in R$ then, $(b,a) \notin R$.

For eg:- The relation $<$ on integers is asymmetric.
 If $a < b$ then it is not $b < a$.
 Therefore $<$ is asymmetric.

Anti-Symmetric	Asymmetric
1. If both relation aRb and bRa are possible then $a=b$ 2. It can be reflexive. 3. Every asymmetric relation is Anti-symmetric.	1. Both relation are not possible. 2. It must be Irreflexive. 3. It cannot be true for every anti-symmetric to be asymmetric.

Q46. Define an equivalence relation. prove that congruence modulo m is an equivalence relation.

A relation R on set A is called equivalence relation if it is reflexive, symmetric and transitive.

To prove: Congruence modulo m is an equivalence relation.

Let a and b be integers define Congruence modulo m .

$$\text{i.e. } a \equiv b \pmod{m}.$$

$$\text{iff and only if, } m \mid (a-b).$$

For reflexive,

for any integer a ,

$$a - a = 0$$

$$\text{since, } m \mid 0,$$

$$a \equiv a \pmod{m}$$

$\therefore$ the relation is reflexive.

for symmetric,

Let $a \equiv b \pmod{m}$

Then,

$$m \mid (a-b)$$

$$m \mid -(b-a)$$

$\therefore (b-a)$ is also divisible by m .

Hence, $b \equiv a \pmod{m}$

Therefore, the relation is symmetric.

For transitive,

Let, $a \equiv b \pmod{m_1}$, $b \equiv c \pmod{m_2}$ then,

$$m_1 \mid a-b, \quad m_2 \mid b-c$$

Now, adding both term,

$$= m_1 \mid (a-b) + m_2 \mid (b-c)$$

$$= m_1 + m_2 \mid a-b+b-c$$

$$= m \mid a-c \quad [m = m_1 + m_2]$$

Hence, $a \equiv c \pmod{m}$ $\therefore$ It is transitive.

Since, the relation is reflexive, symmetric, transitive
 $\therefore$ It is equivalence relation.

Q4. What are equivalence classes? Find the equivalence classes for $\mathbb{Z} \pmod{3}$.

$\rightarrow$ If R is an equivalence relation on A , the equivalence class of an element $a \in A$ is the set of all elements related to a .

$$\text{P.e. } [a] = \{x \in A : x R a\}$$

Equivalence classes of $\mathbb{Z} \pmod{3}$.

For $\pmod{3}$, two integers are equivalent if their difference is divisible by 3.

$$a \equiv b \pmod{3}.$$

There are three equivalence classes, corresponding to the possible remainders 0, 1, 2.

Class of 0:

$$[0] = \{ \dots, -6, -3, 0, 3, 6, \dots \}.$$

$$[0] = \{ 3k : k \in \mathbb{Z} \}.$$

Class of 1:

$$[1] = \{ \dots, -5, -2, 1, 4, 7, \dots \}.$$

$$[1] = \{ 3k+1 : k \in \mathbb{Z} \}.$$

Class of 2:

$$[2] = \{ \dots, -4, -1, 2, 5, 8, \dots \}$$

$$[2] = \{ 3k+2 : k \in \mathbb{Z} \}.$$

$\therefore$ Equivalence classes are $\{ [0], [1], [2] \}$.

Q48.

Define a partially ordered set (poset). provide an example.

A partially ordered set is a ordered pair $(A, \leq)$, where A is non-empty set and $\leq$ is a relation on A which should be reflexive, Anti-symmetric and transitive properties.

for eg:-

Let, a set of positive integers $\mathbb{Z}^+$ under the relation of divisibility ($|$).

For reflexive: Every integer divides itself (eg: $2|2$)
Thus, $a|a$ is true.

For Anti-symmetric: If $a|b$ and $b|a$ then a must equal b .

For transitive: If $a|b$ (so $b = k \cdot a$) and $b|c$ (so $c = m \cdot b$) then $c = m \cdot (k \cdot a) = (m \cdot k) \cdot a$ which is $a|c$.

Hence, $(\mathbb{Z}^+, |)$ is poset.

Q49. What is Totally Ordered set (chain)? How does it differ from a poset?

→ A totally ordered set is a poset in which every pair of element is comparable.

i.e. for any $a, b \in A$,

$$a \leq b \text{ or } b \leq a.$$

For eg:

$$A = \{1, 2, 3, 4\}$$

with usual relation $\leq$ is a totally ordered set because

$$1 \leq 2, 2 \leq 3, 3 \leq 4 \text{ and every pair is comparable.}$$

Poset	Totally ordered set.
1. It must be reflexive, anti-symmetric and transitive which some are comparable & some are incomparable.	2. It must satisfy poset properties where every elements must be comparable.
2. For eg: $A = \{1, 2, 3, 6\}, 1\}$ here, comparability is not required for every pair.	3. For eg: $\{1, 2, 3, 4\}, 1, 2\}$ here, comparability is required for every pair.

Therefore, every chain is a poset but every poset is not necessarily a chain set.

Q50. Explain Hasse Diagrams. Draw a Hasse diagram for the divisibility relation on $\{1, 2, 3, 4, 6, 8, 12, 24\}$

→ Hasse diagram is a graphical representation of the relation of elements of poset with implied upward orientation.

In Hasse diagram.

1. Elements are represented by points (vertices)
2. If $a < b$ then, b is placed above a .

3. Direct connections are drawn only for covering relations.

4. Reflexive, transitive relation are not shown.

For the set,

$A = \{1, 2, 3, 4, 6, 8, 12, 24\}$ under divisibility ($|$).

Here,

$1|2 \quad 1|3$

$2|4 \quad 2|6$

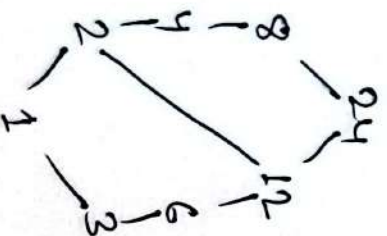
$3|6$

$4|8 \quad 4|12$

$6|12$

$8|24 \quad 12|24$

Hasse diagram is:



Q51.

→

Define Least upper Bound (Supremum) and

Greatest lower bound (Infimum) in a poset.

Least upper bound is the smallest element of all upper bound. If $x \in P$ is any upper bound of A ($a \leq x$ for all $a \in A$) then $y \leq x$. An element $y \in P$ is called least upper bound.

Greatest lower bound is the largest element of all lower bounds. If $y \in P$ is any lower bound of A ($y \leq a$ for all $a \in A$) then $y \leq l$. An element $l \in P$ is called Greatest lower bound.

Q52. Define a lattice. Is every poset a lattice? Justify.

→ A lattice is a poset $(L, \leq)$ in which every pair of element has both a least upper bound and greatest lower bound.

Foreg: for any $a, b \in L$,

$$a \vee b = \text{LUB of } a \text{ and } b.$$

and, $a \wedge b = \text{GLB of } a \text{ and } b.$

must exist for a lattice.

∴ Lattice = poset + LUB + GLB.

Let $L = \{1, 2, 3, 6\}$ under divisibility.

for 2 and 3.

$$2 \vee 3 = 6 \quad \text{because } 6 \text{ is least upper bound.}$$

$$\text{also, } 2 \wedge 3 = 1 \quad \text{because } 1 \text{ is greatest lower bound.}$$

Hence, poset is a lattice.

NO. Every poset is not lattice. because not every poset can have both LUB and GLB.

Q53. How are relations represented using zero-one matrices and directed graph?

→ A relation can be represented using a zero-one matrix or directed graph.

∴ Zero-one matrices representation.

→ Let, $A = \{a_1, a_2, \dots, a_n\}.$

Then, the relation R on A is represented by $m \times m$ matrix.

$$\text{P.E. } MR = [m_{ij}]$$

where,

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, a_j) \in R \\ 0 & \text{if } (a_i, a_j) \notin R \end{cases}$$

$$\{0 \text{ if } (a_i, a_j) \notin R\}.$$

Foreg:

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$.

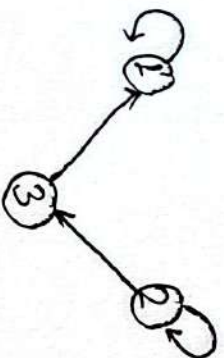
Then,

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

2. Directed Graph Representation.

→ A relation can be represented by directed graph where each element of A is represented by a vertex. If the pair $x \in A, y \in B$ is in the relation, the corresponding nodes are connected by the arcs.

For eg: Let $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (2, 3), (3, 1)\}$ given,



Q54.

Define the inverse of a relation and the composition of two relations.

→ Inverse of a relation is formed by reversing the order of the elements in each ordered pair of the original relation.

Let R be a relation from set A to set B (so, $R \subseteq (A \times B)$). Then,

$$R^{-1} = \{ (b, a) \mid (a, b) \in R \}.$$

Foreg: If $R = \{(1, 2), (2, 1)\}$ then $R^{-1} = \{(2, 1), (1, 2)\}$

Composition of two relations combines them into a single relation by matching intermediate relation elements.

Let R be a relation from set A to set B , let S be a relation from B to set C .

Then, the composition of R and S is $soR = \{(a, c) \mid \text{there exists } a, b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}$

Foregr

If $R = \{(1, 2), (2, 5)\}$ and $S = \{(2, 4), (5, 6)\}$ then
 $soR = \{(1, 4), (2, 6)\}$.

Q55

Explain reflexive closure, symmetric closure and transitive closure of a relation.

→

Closures are used to add the minimum number of ordered pairs required to make a relation with particular properties.

1. Reflexive closure.

→ It is the closure where there is addition of pair to make relation reflexive.

For a relation on A we add, (a, a) for every $a \in A$ that is not already present.

2. Symmetric closure.

→ It is the closure where there is addition of least order pair to make relation symmetric.

For every $(a, b) \in R$, we add $(b, a) \in R$ if it is not already present.

3. Transitive closure.

→ It is the closure where there is addition of least order pair to make relation transitive.

If $(a, b) \in R$ and $(b, c) \in R$ then we must add $(a, c) \in R$ to make it transitive.

Foregr

$R = \{(1, 2), (2, 3)\}$

We need to add at least $(1, 1)$ $R(2, 2)$ for reflexive.
We need to add at least $(2, 1)$ for symmetric.
We need to add at least $(1, 3)$ for transitive.

Q56

→

What is Warshall's algorithm used for? Explain its basic principle.

Warshall's algorithm is used to find the transitive closure of a relation represented by zero-one matrix.

It determines whether there is a path of length one or more from one vertex to another in a directed graph.

Basic principle.

Let M is the adjacency matrix of a directed graph.

Warshall's algorithm checks whether each vertex can be used as intermediate vertex between two other vertices.

$$M_{ij}^{00} = M_{ij}^{00} \vee (M_{ik} \wedge M_{kj}^0)$$

where,

$M_{ij}^{00} = '1'$ means there is a path from i to j .

$m_{ik} = '1'$ means i can reach k .

$m_{kj} = '1'$ means k can reach j .

Therefore, $M_{ik} \wedge M_{kj} = 1$ means i can reach j through k .

Steps:

For each intermediate vertex k .

1. Consider every pair (i, j)
 2. Check whether i can reach k .
 3. Check whether k can reach j .
 4. If both are possible, set $M_{ij}^{00} = 1$.
- $\therefore$ final matrix represents the transitive closure.

It is particularly used for finding reachability between vertices in a directed graph.

Q57.

Define a function. Differentiate between a relation and function.

→ function is a special type of relation in which every element of A is associated with exactly one element of B.

$\forall a \in A, \exists b \in B$ such that $f(a) = b$.

Let A and B two sets. A function f from A to B is $f: A \rightarrow B$

where A = Domains, B = co-domain, $f(a)$ is image of a.

Let $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$ then, $f(x) = 2x$

Then, $f = \{(1, 2), (2, 4), (3, 6)\}$.

Relation	Function.
1. A relation is any subset of $A \times B$.	2. A function is a special type of relation.
2. An element of the domain may have zero, one or multiple images.	2. Every domain element must have exactly one image.
3. It does not necessarily assign an output to every domain element.	3. must assign exactly one output to every domain element.

Q58.

→ Explain injective (one to one) surjective (onto) and bijective functions with examples.

A function $f: A \rightarrow B$ is injective if different elements of A have different images in B. is called injective.

$$f(a) = f(b)$$

$$\therefore a = b$$

For eg:

$$\text{Let } f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$$

$$\text{let } f(a) = f(b)$$

$$\text{or, } 2a+1 = 2b+1$$

$$\text{or } 2a = 2b$$

$$\therefore a = b$$

$\therefore f(x) = 2x+1$ is injective.

2. Surjective function.

$\rightarrow$ A function $f: A \rightarrow B$ is surjective if every element of the codomain B has at least one preimage in A .

i.e. $\forall b \in B, \exists a \in A$ such that $f(a) = b$.

For eg:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x+1. \text{ for any } y \in \mathbb{R}$$

$$y = 2x+1$$

$$\text{or, } x = \frac{y-1}{2}$$

Since $\frac{y-1}{2} \in \mathbb{R}$ for every y has preimage.

$\therefore f$ is surjective.

3. Bijective function.

$\rightarrow$ A function is bijective if it is both injective and surjective.

For eg: $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x+1$, It is both injective and surjective. Therefore it is bijective.

Q59. Let $f(x) = 2x+3$ and $g(x) = x^2$. Find the compositions.

$$f(g(x)) \text{ and } g(f(x)).$$

$$\rightarrow \text{Given } f(x) = 2x+3, g(x) = x^2$$

Now,

$$f(g(x)) = f(x^2) \\ = 2(x^2) + 3$$

$$\therefore f(g(x)) = 2x^2 + 3$$

$$\text{And, } g(f(x)) = g(2x+3) \\ = (2x+3)^2$$

$$\therefore g(f(x)) = 4x^2 + 12x + 9.$$

Q80. Explain the concept of the inverse of a function.

Find the inverse of $f(x) = \frac{x-2}{3}$.

→ An inverse function $f^{-1}: B \rightarrow A$ reverses the mapping of f .

If $f(a) = b$ then $f^{-1}(b) = a$.

The inverse satisfies $f^{-1}(f(x)) = x$ or $f(f^{-1}(x)) = x$.

A function has an inverse function if and only if it is bijective between specified domain and co-domain.

Given, $f(x) = \frac{x-2}{3}$.

Let $y = \frac{x-2}{3}$ then, Interchanging x and y we get,

$$\text{or } x = \frac{y-2}{3}$$

$$\text{or } 3x = y - 2$$

$$\text{or } y = 3x + 2$$

Hence, $f^{-1}(x) = 3x + 2$.

Q81) Define ceiling and floor functions. Find $[3.1]$ and $[2.9]$.

→ The ceiling function of a real number x is the smallest integer greater than or equal to x .

$$[x] - 1 < x \leq [x]$$

$$\text{Foreg: } [3.2] = 4.$$

The floor function of a real number x is the greatest integer less than or equal to x .

$$\text{Fc. } [x] \leq x < [x] + 1$$

$$\text{Foreg: } [3.7] = 3.$$

Now,

Given,

$$[-3.1] \bullet$$

$$= -4 \leq -3.1 \leq -3$$

$$= -4$$

which is greatest integer less than or equal to -3.1

$$\therefore [-3.1] = -4$$

$$[2.9]$$

$$= 2 \leq 2.9 \leq 3$$

$$= 3$$

which is smallest integer greater than or equal to 2.9 .

$$\therefore [2.9] = 3.$$

Q62. Discuss the application of relations in Relational Database Management Systems.

A Relational Database Management System stores data in the form of tables, where tables consists of rows and columns.

Applications of relations in RDBMS:-

1. Representing data.

→ A table can be viewed as a relation. Each row represents a tuple and the entire table represents a relation.

Student ID	Name	Course.
101	Ram	BCA
102	Sita	BIT
103	Ham	BCA.

2. primary key.

→ A primary key uniquely identifies each record.
For eg: Student ID can be used as primary key.

3. Querying data.

→ Relations allow operations such as: selection, projection, join, union, intersection, difference. These operations are fundamentals to relational algebra and are used in database queries.

Q63. Explain how hash function work and their relation to discrete mathematics

→ A hash function is a function that converts an input of a arbitrary size into fixed size value called hash value, hash code.

$$f.e. h: U \rightarrow \{0, 1, \dots, m-1\}$$

where, U = set of possible inputs.

m = number of possible hash locations.

$$h(x) = \text{hash value of input } x.$$

Suppose, we have a hash table of size 10.

A simple hash function can be:

$$h(k) = k \bmod 10.$$

for eg: for key 37.

$$h(37) = 37 \bmod 10 = 7$$

∴ key 37 is stored at position 7.

A collision occurs when two different inputs produce same hash value.

For eg:

$$h(27) = 27 \bmod 10 = 7$$

Therefore, 27 and 37 have same hash location.

Since there may be more possible inputs than hash value, two different inputs must sometimes map to same hash value. (Pigeonhole).

Thus, Hash function provide an efficient way to store and retrieve data.

Unit 4: Mathematical Reasoning and Proof Techniques.

Q64. $\rightarrow$

State the principle of mathematical induction.

The principle of mathematical induction states that the statement $P(n)$ is true for all positive integers (n) where $n \geq 1$. If the following two conditions are met:

1. Base case.

$\rightarrow$ In this step, we verify that the statement is true for initial value typically $n=1$.
i.e. $P(1)$ is true.

2. Inductive step

$\rightarrow$ In this step, we prove that if the statement hold for an arbitrary positive integer k , then it must also hold for next integer $k+1$.
i.e. $\forall k, P(k) \rightarrow P(k+1)$ is true.

Q65.

Using mathematical induction, prove that $1+2+3+\dots+n = \frac{n(n+1)}{2}$.

$\Rightarrow$ Let $P(n)$ be the proposition that the sum of the first n positive integers is $\frac{n(n+1)}{2}$.

Now,

Base step:

Let $P(1)$ is true for given proposition.

$$P(1) = \frac{1(1+1)}{2} = 1 \text{ which is true.}$$

Induction hypothesis:

Let $P(k)$ holds for an arbitrary positive integers k .
i.e.

$$1+2+3+\dots+k = \frac{k(k+1)}{2}$$

Inductive step:

Here, we have to show that $P(k+1)$ is true under induction hypothesis.

Here,

$$1+2+3+\dots+k+(k+1) = \frac{(k+1)(k+1+1)}{2} \text{ is also true.}$$

From induction hypothesis, we have,

$$1+2+3+\dots+k = \frac{k(k+1)}{2}.$$

Adding both sides by $k+1$.

$$\begin{aligned} 1+2+3+\dots+k+k+1 &= \frac{k(k+1)}{2} + k+1 \\ &= \frac{k(k+1) + 2k+2}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

Therefore, $P(k+1)$ is also true under induction hypothesis. This completes the proof.

Q6. prove by induction that $1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$

→ Let $P(n)$ be the proposition that the sum of square of first n positive integer is $\frac{n(n+1)(2n+1)}{6}$.

Now,

Base case:

Let $P(1)$ is true for given proposition.

$$\begin{aligned} P(1) &= \frac{1(1+1)(2(1)+1)}{6} \\ &= \frac{2 \times 3}{6} \\ \therefore P(1) = 1 \text{ is true.} \end{aligned}$$

Inductive hypothesis:

Let $P(k)$ holds for an arbitrary positive integer k .

$$1^2+2^2+3^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}.$$

Inductive step:

Here, we need to prove that $P(k+1)$ is true under inductive hypothesis.

Here,

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

Taking LHS,

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$$

From the induction hypothesis,

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6},$$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= (k+1) \left[\frac{k(2k+1) + 6(k+1)}{6} \right]$$

$$= (k+1) \left[\frac{2k^2 + k + 6k + 6}{6} \right]$$

$$= (k+1) \left[\frac{2k^2 + 7k + 6}{6} \right]$$

$$= (k+1) \left[\frac{2k^2 + 4k + 3k + 6}{6} \right]$$

$$= (k+1) \left[\frac{(2k+3)(k+2)}{6} \right]$$

$\therefore P(k+1)$ is true under the inductive hypothesis
 $P(k)$ true. This completes proof.

Q67.

→ prove using mathematical induction that the sum of the first n odd positive integers is n^2 .

Given, $1 + 3 + 5 + \dots + (2n-1) = n^2$

Let $P(n)$ be the proposition that sum of the first n positive odd integers is n^2 .

Now,

Base case:-

Let $P(1)$ is true for given proposition.

$$P(1) = (1)^2 \\ = 1 \text{ which is true.}$$

Induction Hypothesis:

Let $P(k)$ is true where $k \in \mathbb{Z}$.
i.e.

$$P(k) = 1 + 3 + 5 + \dots + (2k-1) = k^2$$

Inductive step:

Now, we need to show $P(k+1)$ is true under induction hypothesis.

$$P(k+1) = 1 + 3 + 5 + \dots + (2k-1) + (2k+1) = (k+1)^2$$

Taking LHS,

$$1 + 3 + 5 + \dots + (2k-1) + 2k + 1$$

$$= k^2 + 2k + 1 \quad [\because \text{From the induction hypothesis}]$$
$$= (k+1)^2 = \text{RHS}.$$

$\therefore P(k+1)$ is true under $P(k)$ is true. This completely proof.

Q68. prove by induction that $n^3 - n$ is divisible by 3 for all positive integers n .

$\rightarrow$ Let $P(n)$ be the proposition for $n^3 - n$ is divisible by 3.

Base step:

Let $P(n)$ is true for $n=1$

$$\text{i.e. } P(1) = 1^3 - 1$$
$$= 0 \text{ which is divisible by 3.}$$

Induction hypothesis:

Let $P(n)$ is true for $n=k$

i.e. $P(k)$, $k^3 - k$ is divisible by 3.

Inductive step:

Here, we need to show $P(k+1)$ is true under induction hypothesis.

Now,

$$P(k+1) = (k+1)^3 - (k+1)$$
$$= k^3 + 3k^2 + 3k + 1 - k - 1$$
$$= k^3 + k + 3k^2 + 3k$$

$$= k^3 + k + 3Ck^2 + k$$

Since, $k^3 + k$ is divisible by 3 under induction hypothesis and $3Ck^2 + k$ is multiple of 3 which is also divisible by 3. Therefore $P(k+1)$ is also true under $P(k)$ is true. Thus it is proved that $n^3 - n$ is divisible by 3 under n is positive integer.

Q69.

Use induction to prove that $2^n > n$ for all non negative integers.

Let $P(n)$ be the propositional for $2^n > n$ for all non-negative integers.

Base case:

Let $P(n)$ is true for $n=0$.

$$P(0) = 2^0 > 0$$

$= 1 > 0$ which is true.

Inductive Hypothesis:-

Let $P(k)$ be true for an arbitrary non-negative integer k .

$$\text{ie. } P(k) = 2^k > k.$$

Inductive step:

Here, we need to show that $P(k+1)$ is true for $n = k+1$

$$\text{ie. } P(k+1) = 2^{k+1} > k+1$$

$$= 2^k \cdot 2 > 2^{k+1}$$

$$= 2 \cdot 2^k \geq 2k$$

(Substituting Inductive Hypothesis $2^k > k$)

$$2 \cdot 2^k > k + k$$

Since k is non-negative integer ($k \geq 0$).

If $k \geq 1$ then $k+k \geq k+1$

If $k=0$, we can check $n=1$ directly: $2^1 > 1$ which is true.

Thus for all $k \geq 0$ we have $2k \geq k+1$.

Therefore, $2^{k+1} > k+1$ when it is true.

Thus P_{k+1} is true for induction hypothesis when prove $2^n > n$ for all non-negative integers.

Q36.

What is strong mathematical induction? when is it preferred over regular induction?

→

Strong mathematical induction is a proof technique where the statement $P(n)$ is true for all integers by assuming it holds for all preceding ~~rather~~ than just immediate previous step.

Base case: let $P(n)$ is true for $n \leq b$ and sometimes multiple base cases like $n \leq b, b+1, b+2$ are needed.

Inductive hypothesis: let $P(j)$ are true for all $j = 1, 2, 3, \dots, k$ where $j \leq k$.

Inductive step: P_{k+1} must be true based on that assumption.

Strong induction is preferred when the truth of the next step (n+1) depends on multiple previous states or an arbitrary smaller state rather than just the single step right before it (n). It is required when to prove the properties of prime numbers, recursive sequences, divide and conquer algorithm etc.

Q37.

State the well-ordering principle. How does it relate to mathematical induction?

→

The well-ordering principle states every non-empty subset of the set of positive integers has a least element.

The well-ordering principle and mathematical induction are equivalent principles.

This means that each can be used to prove the other.

In mathematical induction, we prove:-

Base case: The statement is true for initial value.

Inductive step: If statement is true for $n=k$. Then it is true for $n=k+1$.

In well-ordering principle:-

If a statement were false for some positive integer, the set of all counter examples would have least element. using inductive hypothesis, we can show that this smallest counter example cannot exist.

Therefore, well-ordering principle is one of the fundamental principles underlying mathematical induction.

Q72. What are recursive definitions? Define the factorial function recursively.

→ A recursive definitions define an object in terms of base case which gives initial value and recursive case which defines new values using previously defined values. It must have a base case so that the process eventually terminates.

Let the factorial of a non-negative integer n is denoted by $n!$.

Its recursive definition is $0! = 1$. and for $n \geq 1$

$$n! = n(n-1)!$$

$$\text{Foregr } 5! = 5 \times (4!)$$

$$= 5 \times (4! \times (3!))$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 5 \times 4 \times 3 \times 2 \times 1 \times 1$$

$$= 5 \times 4 \times 3 \times 2 \times 1 \times 0!$$

$$\therefore 5! = 120 \quad [\because 0! = 1]$$

Q73. Give a recursive definition for the Fibonacci Sequence.

→ The Fibonacci sequence is recursively defined as

$$F_0 = 0, F_1 = 1 \text{ [Base case]}$$

and for $n \geq 2$,

$$F_n = F_{n-1} + F_{n-2} \text{ [Recursive case]}$$

First few Fibonacci numbers,

$$F_0 = 0$$

$$F_1 = 1$$

$$F_2 = F_0 + F_1 = 1$$

$$F_3 = F_1 + F_2 = 2$$

$$F_4 = F_2 + F_3 = 3$$

$$F_5 = F_3 + F_4 = 5$$

Thus, Fibonacci series are: 0, 1, 1, 2, 3, 5, ...

Q74. Prove by induction that the n th Fibonacci no. F_n is less than 2^n .

→ Need to prove, $F_n < 2^n$.

Let $P(n)$ be the proposition for $F_n < 2^n$ for all $n \geq 1$.

Base case;

Let $P(n)$ is true for $n=1$

$$F_1 = 1, 2^1 = 2$$

$\therefore 1 < 2$ which is true.

Inductive hypothesis:-

Let $P(k)$ is true for all $n=k$

$$\text{i.e. } P(k) = F_k < 2^k$$

and let $P(k+1)$ is true for $n=k+1$

$$\text{i.e. } P(k+1) = F_{k+1} < 2^{k+1}.$$

Inductive step.

We need to prove that P_{k+1} is true under inductive hypothesis.

We have,

$$F_{k+1} = F_k + F_{k-1}$$

using induction hypothesis,

$$F_{k+1} < 2^k + 2^{k-1}$$

since,

$$\begin{aligned} 2^k + 2^{k-1} &= 2^{k-1}(2+1) \\ &= 3(2^{k-1}) < 4(2^{k-1}) \\ &= 2^{k+1} \end{aligned}$$

We get,

$$F_{k+1} < 2^{k+1}$$

$\therefore$ The statement is true for $k+1$

Thus, By mathematical induction, $F_n < 2^n$ for all $n \geq 1$

Q75.

Define sets recursively. Give a recursive definition of the set of even positive integers.

A set can be defined recursively using basis which specify the initial elements of the set, recursive case which specify how to generate new elements from existing elements and closure which include only the elements obtained using basis and recursive case.

Let E be the set of all positive even integers.

Base case;

Let $P(n)$ is true for $n \geq 2$.

i.e. $2 \in E$.

Recursive case;

If let $P(n)$ is true for $n \geq k$

i.e. $k \in E$.

Then,

$k+2 \in E$

$\therefore E = \{2, 4, 6, 8, \dots\}$.

A complete recursive definition is:

$$\{ 2 \in E \\ n \in E \rightarrow n+2 \in E \}$$

Q76.

→

Explain Structural Induction. Provide an example of its application on trees or lists.

Structural Induction is a proof technique used to prove properties of recursively defined structures such as, trees, lists, string, recursive data structure.

The steps of structural induction:

1. Base case: prove the properties for the simplest structure.
2. Inductive step: Assume the property holds for smaller structures and prove it for larger structures.

Therefore, the property holds for all recursively constructed structures.

For eg:

Suppose a binary tree is recursively defined as single node is a binary tree and if T_1 and T_2 are binary trees then new roots with T_1 and T_2 as its subtrees is also binary tree.

We can prove:

Every binary tree with n nodes has exactly $n-1$ edges.

Base case:

For a tree with one node.

$n=1$, It has no edges as $n-1=1-1=0$. which is true.

Inductive step.

Let two smaller binary tree T_1 and T_2 satisfy the property.

Let T_1 has n_1 nodes and $n_1 - 1$ edges.

and T_2 has n_2 nodes and $n_2 - 1$ edges

When we create new root and attach T_1 and T_2 to it, the resulting tree has:

$$n = 1 + n_1 + n_2 \text{ nodes.}$$

It number of edges are:

$$\begin{aligned} &= (n_1 - 1) + (n_2 - 1) + 2 \\ &= n_1 + n_2 \end{aligned}$$

$$\text{and } n - 1 = (1 + n_1 + n_2) - 1 = n_1 + n_2$$

$$\therefore \text{Number of edges} = n - 1$$

Thus, By structural induction every binary nodes n has exactly $n - 1$ edges.

Q37.

→ prove by induction that $5^n - 1$ is divisible by 4 for all integers $n \geq 1$.

Let $P(n)$ be the proposition that $5^n - 1$ is divisible by 4 for $n \geq 1$.

Base case:

Let $P(n)$ is true for $n = 1$

$$\text{i.e. } P(1) = 5^1 - 1 = 4 \text{ which is divisible by 4.}$$

Inductive Hypothesis:

Let $P(k)$ is true for $n = k$.

$$\text{i.e. } P(k) = 5^k - 1 \text{ is divisible by 4.}$$

Inductive step:

Here we need to prove $P(k+1)$ is true under Inductive Hypothesis.

$$\begin{aligned} P(k+1) &= 5^{k+1} - 1 \\ &= 5^k \cdot 5^1 - 1 \\ &= 5^k \times 5^1 - 5 + 4 \\ &= 5(5^k - 1) + 4 \end{aligned}$$

Here, $5^k - 1$ is divisible by 4 from hypothesis. So, $5^k - 1$ is divisible by 4 for all $n \geq 1$.

Q78. Prove that $1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

→ Let $P(n)$ be the proposition of $1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$.

Base case:

Let $P(n)$ is true for $n=1$

$$L.H.S = 1.2 = 2$$

$$R.H.S = \frac{1(1+1)(1+2)}{3} = 2$$

∴ LHS = RHS. which is true

Inductive Hypothesis:

Let $P(n)$ is true for $n=k$.

$$P.O. \quad P(k) = 1.2 + 2.3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

Inductive step:

Let $P(k+1)$ is true under inductive hypothesis.

$$P.O. \quad P(k+1) = 1.2 + 2.3 + \dots + k(k+1) + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

Now, Taking LHS.

$$1.2 + 2.3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \quad [\text{from inductive hypothesis}]$$

$$= \frac{k(k+1)(k+2) + 3\{(k+1)(k+2)\}}{3}$$

$$= \frac{(k+1)(k+2)\{k+3\}}{3} = R.H.S.$$

∴ $P(k+1)$ is true under induction hypothesis.

Therefore, $1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ is true for all $n \geq 1$.

Discuss how induction is applied in proving the correctness of loop invariants in algorithms.

→

A loop invariant is a condition that remains true before and after every iteration of a loop. Mathematical induction is closely related to proving the correctness of a loop invariant.

To establish correctness using induction, proof consists three structural properties:

1. Initialization (Base case)

→ we first prove that the loop invariant is true before first iteration of the loop. This corresponds to the base case in mathematical induction.

2. Maintenance (Inductive Step)

→ Let the loop invariant is true before an iteration. We then show that after executing the loop body the invariant is still true. This corresponds to the inductive hypothesis and inductive step.

If the invariant is true before iteration k and the loop body preserves it then it is also true before iteration $k+1$.

3. Termination.

→ When the loop is terminated, the invariant is still true. It can be used to prove that the algorithm produces the required result combined with loop invariant termination condition.

For example-

Let an algorithm that calculates the sum of the first n positive integer.

Algorithm:-

Sum = 0

for $i = 1$ to n :

Sum = Sum + i

A suitable loop invariant is:

$$\text{Sum} = 1 + 2 + \dots + (i^0 - 1)$$

Initialization: Before the first iteration, $i = 1$ and $\text{sum} = 0$. Thus, the invariant is true.

Maintenance:

$$\text{Let } \text{sum} = 1 + 2 + \dots + (i^0 - 1),$$

before the next iteration, the algorithm adds,

$$\text{sum} += 1 + 2 + \dots + (i^0 - 1) + i^0.$$

Therefore, the invariant remains true.

Termination:

When $i = n+1$.

$$\text{sum} = 1 + 2 + \dots + (i^0 - 1)$$

$$\text{sum} = 1 + 2 + \dots + (n+1-1)$$

$$\text{sum} = 1 + 2 + \dots + n$$

Thus, the algorithm correctly calculates the sum.

Hence, mathematical induction provides the logical foundation for proving that a loop invariant remains true throughout all iterations of an algorithm.

Q80.

→

Prove that a set with n elements has exactly 2^n subsets using mathematical induction.

Let $P(n)$ be the proposition of a set containing n elements having exactly 2^n subsets.

Base case:

Let $P(n)$ is true for $n=0$.

Pf. The set $S = \{\emptyset\}$ for $n=0$ (empty set)

$$P(0) = 2^0 = 2^0 = 1 \text{ which is true for } n=0.$$

Alternatively,

$P(n)$ is true for $n=1$, a set $\{a\}$ has two subsets

$$S = \{\emptyset, \{a\}\}$$

$P(1) = 2^1 = 2$ which is true.

Inductive hypothesis:

Let that a set containing k elements has exactly 2^k subsets. $S = \{a_1, a_2, \dots, a_k\} = 2^k$.

Inductive step:

Let a set containing $k+1$ elements.

$$S = \{a_1, a_2, \dots, a_k, a_{k+1}\}$$

Then,

Separate its subsets into two groups:

1. subsets not containing a_{k+1} .
2. subsets containing a_{k+1} .

The first group contain exactly 2^k subset by the induction hypothesis.

For every subset not containing a_{k+1} , we can create another subset by adding a_{k+1} . Therefore, the second group also contains 2^k subsets.

Hence, total number of subsets is:

$$\begin{aligned} &= 2^k + 2^k \\ &= 2 \cdot 2^k \\ &= 2^{k+1} \end{aligned}$$

Thus, the statement is true for $k+1$.

Therefore, by the principle of mathematical induction, a set with n elements has exactly 2^n subsets.

Q81. prove that $n! \geq 2^n$ for all integers $n \geq 4$.

→ Let $P(n)$ be the proposition of $n! \geq 2^n$ for all integers $n \geq 4$.

Base case:

Let $P(n)$ is true for $n=4$.

$$P(4) = 4! = 4 \times 3 \times 2 \times 1 = 24$$

and,

$$P(4) = 2^4 = 16$$

$\therefore 24 > 16$ which is true for $n=4$.

Inductive Hypothesis:

Let $P(n)$ is true for some integer $k \geq 4$.

i.e. $P(k) = k! > 2^k$.

Inductive step:

Here, we need to prove $P(k+1)$ is true under inductive hypothesis.

i.e. $P(k+1) = (k+1)! > 2^{k+1}$

Taking LHS,

$$= (k+1)!$$

$$= (k+1)k!$$

Using induction hypothesis,

$$(k+1)! > (k+1)2^k.$$

Since $k \geq 4$.

$$(k+1) \geq 5 > 2$$

$$\therefore (k+1)2^k > 2^k \cdot 2$$

$$= 2^{k+1}$$

Hence, $(k+1)! > 2^{k+1}$

Therefore, By mathematical induction,

$$n! > 2^n \text{ for every integer } n \geq 4.$$

Q82. Explain the concept of basis step and inductive step in detail.

→

Mathematical induction is method used to prove that a statement is true for all positive integer or for all integers from a specified initial point. It includes two steps:

1. Basis step.

→ It is called base case that verifies that the statement is true for the first value of n .

Let $P(n)$ for all $n \geq n_0$.

$P(n_0)$ should be true.

for eg:

$$1+2+\dots+n = \frac{n(n+1)}{2}$$

Let $P(n)$ is true for $n=1$

$$P(1) = \frac{1(1+1)}{2} = 1 \text{ which is true.}$$

2. Inductive step.

→ In this step, we assume that the statement is true for an arbitrary element k . This assumption is called induction hypothesis.

i.e. $P(k)$ is true for $n=k$.

Then, we prove that $P(k+1)$ is true using that hypothesis.

for eg:

$$P(k) = 1+2+\dots+k = \frac{k(k+1)}{2}$$

To prove:

$$P(k+1) = 1+2+\dots+k+(k+1) = \frac{(k+1)(k+2)}{2}$$

Taking LHS,

$$= 1+2+\dots+k+(k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) \quad [\text{using hypothesis}]$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)^2}{2} (k+2)$$

∴ $P(k+1)$ is true under hypothesis.

Thus, the process is represented as,

$$P(n_0) \Rightarrow P(n_0+1) \Rightarrow P(n_0+2) \dots$$

Therefore, the statement is true for every integer $n \geq n_0$.

Q83.

Outline the common fallacies or mistakes made when writing inductive proofs.

Several common mistakes make a mathematical induction proof invalid.

1. Missing the basis step.

→ A proof may only show that $P(k) \rightarrow P(k+1)$ without proving that $P(n_0)$ is actually true. This is incomplete because there is no initial point.

2. Proving the wrong base case.

→ The proof must begin at the value specified in the statement.

Foreg: If a starting base is 1 but we take wrong integer which fails to ground the domino effect for the target domain.

3. Proving the wrong statement for $k+1$.

→ The induction step must establish exactly the original statement with n replaced by $k+1$.

4. Confusion for k and $k+1$.

→ Care must be taken when substituting values into formulas.

5. Failing to use induction hypothesis.

→ The induction hypothesis must be used to prove the statement for $k+1$.

Q84.

prove by induction that $\sum_{i=1}^n ar^{i-1} = \frac{a(r^n - 1)}{r - 1}$ for $r \neq 1$.

→ Let $P(n)$ be the proposition for $\sum_{i=1}^n ar^{i-1} = \frac{a(r^n - 1)}{r - 1}$.

Base case:-

Let $P(n)$ is true for $n=1$.

$$\text{ie } P(1) \Rightarrow \sum_{i=1}^1 ar^{i-1} = ar^0 = a$$

and,

$$= \frac{a(r^1-1)}{(r-1)} = \frac{a(\cancel{r^1}-1)}{\cancel{a(r-1)}} = a$$

$\therefore$ LHS = RHS which is true for $n=1$.

Induction hypothesis:

Let $P(n)$ is true for $n \leq k$.

$$\text{i.e. } P(k) = \sum_{i=1}^k ar^{i-1} = \frac{a(r^k-1)}{r-1}.$$

Induction step:-

here, we need to prove that $P(n)$ is true for $n=k+1$.

$$\text{P.e. } P(k+1) = \sum_{i=1}^{k+1} ar^{i-1} = \frac{a(r^{k+1}-1)}{(r-1)}$$

Now, Taking LHS,

$$\begin{aligned} \sum_{i=1}^{k+1} ar^{i-1} &= \sum_{i=1}^k ar^{i-1} + ar^{(k+1)-1} \\ &= \frac{a(r^k-1)}{r-1} + ar^k \quad [\because \text{Induction hypothesis}] \\ &= \frac{a(r^k-1) + ar^k(r-1)}{r-1} \\ &= \frac{ar^k - a + ar^{k+1} - ar^k}{r-1} \\ &= \frac{ar^{k+1} - a}{r-1} \\ &= \frac{a(r^{k+1}-1)}{r-1} = \text{R.H.S} \end{aligned}$$

$\therefore P(k+1)$ is true.

Since, the statement is true for $n=1$, and its truth for $n=k$ implies its truth for $n=k+1$ by the mathematical induction. Therefore, $\sum_{i=1}^n ar^{i-1} = \frac{a(r^n-1)}{r-1}$.

Unit-5: Combinatorics and Counting Principles.

Q85. State the fundamental counting principles. The rule of sum and the rule of product.

→ The rule of sum states that if an event E can occur in 'm' ways and another event F can occur in 'n' ways and the two events cannot be performed simultaneously then there are $m+n$ ways that can occur one of the two events.

The rule of product states that if an event E can occur in 'm' ways and independent of this event, if another event F occur in 'n' ways then two events can occur simultaneously in $m \times n$ ways.

Q86. A student can choose a computer project from 20 software lists or 15 hardware lists. How many choices do they have?

→ Since, the student can choose either software project or hardware project. The choices can be $20+15=35$.

Q87. How many different bit strings of length 8 are there?

→ A bit can be either 0 or 1, so each position has 2 choices.

for 8 positions: There are $2^8 = 256$ bit strings in the length '8'.

Q88. Explain the concept of permutations. state the formula for $P(n,r)$.

→ A permutation is an arrangement of objects in which order matters. Unlike combinations, the order of selection matters crucially in permutations. For example, Arranging A and B gives: AB, BA. which are 2 different permutations.

The formula for permutations of n objects taken r at a time is $P(n, r) = \frac{n!}{(n-r)!}$

where, n is the total number of unique objects available, r is the number of objects being selected or arranged, constraint $0 \leq r \leq n$.

Q89. Explain combinations. state the formula for $C(n, r)$

→ A combinations is a selection of objects in which order does not matter.

foregr selecting A and B is the same as selecting B and A.

selecting r items from the total group of ' n ' unique item is $C(n, r) = \frac{n!}{r!(n-r)!}$

where, n is the total number of items in the group, r is the number of items being chosen.

Q90. How many ways can a committee of 5 be chosen from a group of 12 people?

→ Here, the selection of people does not matter the order of a people. so we use combination. As we have to select 5 people from group of 12 people.

$$\begin{aligned} \text{i.e. } C(12, 5) &= \frac{12!}{5!(12-5)!} \\ &= \frac{12!}{5! \times 7!} \\ &= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7!}{5 \times 4 \times 3 \times 2 \times 1 \times 7!} \\ &= 792 \end{aligned}$$

∴ There are 792 ways to select 5 people from 12.

Q91. How many permutations of the Letter ABCDEFGH contain the string ABC?

→ There are 8 letters in ABCDEFGH.

Since ABC must remain together and in that order ABC is treated as single block.

Now, we have 6 objects to arrange.

[ABC], D, E, F, G, H.

Therefore, there are $6!$ ways to arrange these letters.

i.e. 720 ways.

Q92. State the simple pigeonhole principle. Give a real-world example.

→ Simple pigeonhole principle states that if $n+1$ objects are placed into n boxes then at least one box must contain at least two objects.

For eg:

There are 13 people but only 12 months in a year.

Therefore, at least two people must have their birthdays in same month.

Q93. State the generalized pigeonhole principle.

→ The generalized pigeonhole principle states that if ' N ' objects are placed into ' k ' boxes then there is at least one box containing at least $\lceil \frac{N}{k} \rceil$ objects.

Here, $\lceil x \rceil$ represents the ceiling function (which rounds x up to nearest integer).

For eg:

If 25 students are divided among 6 classes then,

Each class gets at least $= \lceil \frac{25}{6} \rceil = \lceil 4.167 \rceil = 5$.

Therefore, at least one class must have 5 or more students.

Q94. Show that among any group of 367 people, there must be at least two with same birthday.

→ There are at most 365 possible birthdays in a year. (Ignoring February 29). We have 367 people and possible 365 birthdays.

By the pigeonhole principle,

Number of people > Number of possible birthdays.

$$367 > 365$$

If each person had different birthday then there could be at most 365 people. But there are 367 people. Therefore, at least two people must have the same birthday.

Using Generalized principle,

$$\left\lceil \frac{367}{365} \right\rceil = \lceil 1.0054 \rceil = 2$$

Hence, at least two people have the same birthday.

Q95. What is the minimum number of students required in a discrete math class to guarantee that at least 6 receive the same grade (assuming grades are A, B, C, D, F)?

→ There are 5 possible grades i.e. A, B, C, D, F.

Using pigeonhole principle,

To avoid having 6 students with same grade, we can have at most 5 students in each grade.

Therefore, there can be $5 \times 5 = 25$ students with possible grades. i.e. 5A, 5B, 5C, 5D, 5F. The next student i.e. 26th student must make one grade have at least 6 students. Hence, there can be 26 minimum students.

Q96. State the Inclusion-Exclusion principle for two sets and three sets.

→ The Inclusion-Exclusion principle is a counting technique used to find a number of elements in the union of multiple sets. It prevents double counting elements that belong to more than one set.

Inclusion-Exclusion for two sets.

For any two finite sets A and B, the number of elements in their union is:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

$|A \cup B|$: Total elements in either set A or set B.

$|A| + |B|$: sum of elements in each set (this counts overlapping element twice).

$|A \cap B|$: subtracts the overlapping elements to correct the double counting.

For three sets.

For any three finite sets A, B and C, the number of elements in their union is:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|.$$

$|A| + |B| + |C|$: counts all elements in the sets (overlap are counted multiple times).

$|A \cap B|, |A \cap C|, |B \cap C|$: Need to subtracts the elements shared by two sets.

$|A \cap B \cap C|$: Adds back the elements shared by all three sets.

Q97. In a class of 50 students, 30 study Python, 25 study Java and 10 study both. How many study neither?

→ Given: $|U| = 50$
 $|P| = 30$ [where P is student studying python]
 $|J| = 25$ [where J is student studying Java]
 $|P \cap J| = 10$

Using Inclusion-Exclusion principle,

$$\begin{aligned} |P \cup J| &= |P| + |J| - |P \cap J| \\ &= 30 + 25 - 10 \\ &= 45 \end{aligned}$$

Now,

$$\begin{aligned} |\overline{P \cup J}| &= |U| - |P \cup J| \\ &= 50 - 45 \end{aligned}$$

$$\therefore |\overline{P \cup J}| = 5$$

$\therefore$ 5 students neither study python nor Java.

Q98. Explain permutations with repetitions. How many distinct arrangements are there of the word "MISSISSIPPI"?

→ When some objects are identical, the number of distinct permutation is $\frac{n!}{n_1! \cdot n_2! \cdot n_3! \dots n_k!}$

where, n is the total number of objects and $n_1, n_2, \dots, n_k$ is the frequencies of repeated objects.

Given:

MISSISSIPPI.

It has 11 letter and I is repeated for 4 times, s is for 4 times, p is for 2 times.

Therefore,

$$\begin{aligned}\text{Number of arrangements} &= \frac{11!}{4! \cdot 4! \cdot 2!} \\ &= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!}{4! \times 4! \times 2!} \\ &= 34650\end{aligned}$$

∴ There are 34650 ways to arrange letter MISSISSIPPI.

Q99. What are circular permutations? How many ways can 6 people sit around a circular table?

→ Circular permutations is an arrangement of object around a circle where arrangement that differ only by rotation are considered the same.

For n distinct people around a circular table $(n-1)!$

For 6 people, There are $(n-1)! = (6-1)!$ ways

$$\begin{aligned}&= 5! \\ &= 5 \times 4 \times 3 \times 2 \times 1 \\ &= 120 \text{ ways.}\end{aligned}$$

∴ There are 120 ways to sit around a circular table for 6 people.

Q100. State the Binomial Theorem and explain Pascal's Identity.

→ Binomial Theorem states that the expansion for any non-negative integer power of a binomial $(x+y)^n$ for any positive integer n ,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} \cdot y^k.$$

Where, $\binom{n}{k}$ represents $C(n, k) = \frac{n!}{k!(n-k)!}$.

The expansion contains exactly $(n+1)$ terms and the power of x decrease from n to 0 while power of y increase from 0 to n .

Pascal's Identity is a fundamental formula that describes the relationship between adjacent binomial coefficients. It states that for any positive integers n and k where $1 \leq k \leq n$:

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

It means that each entry in Pascal's triangle is obtained by adding two number/entries directly above it.

Q101. Find the coefficient of x^4y^3 in the expansion $(x+y)^7$.

→ Given, $(x+y)^7$

using binomial expansion,

$$(x+y)^7 = \sum_{r=0}^7 \binom{7}{r} x^{n-r} y^r$$

$$\text{Here, } n-r = 7-3 = 4 \\ r = 3$$

$$\begin{aligned} \text{Now, the coefficient } \binom{7}{r} &= \binom{7}{3} = \frac{7!}{3!4!} \\ &= \frac{7 \times 6 \times 5 \times 4!}{3 \times 2 \times 4!} \\ &= 35 \end{aligned}$$

∴ the coefficient of $x^4y^3 = 35$.

Q102. Explain combinations with repetition (stars and bars method).

→ The stars and bars method is a graphical technique used in combinatorics to solve problems where you need to place identical items into distinct bins which is the definition of combination with repetition. It allows an object to be selected more than once.

If 'n' identical objects among r distinct group is distributed allowing zero objects in a group then number of ways is $\binom{n+r-1}{r-1}$

Exeg:

Let $k=5$ pieces of fruit and sells $n=3$ types of fruit.

using the formula, the total number of ways to choose fruit is

$$\binom{5+3-1}{3} = \binom{7}{3} = \frac{7!}{3!4!} = 21 \text{ ways.}$$

Q103. How many ways are there to distribute 10 identical apples to 4 children.

→ we have,

$$n=10, r=4$$

using stars and bars,

$$\text{Number of ways} = \binom{10+4-1}{4-1}$$

$$= \binom{13}{3}$$

$$= \frac{13!}{3!10!}$$

$$= \frac{13 \times 12 \times 11 \times 10!}{3 \times 2 \times 10!}$$

$$= 286$$

∴ There are 286 ways to distribute 10 identical apples to 4 children.

Q104. What are derangements? Give the formula for calculating the number of derangement of n objects.

→ A derangement is a permutation in which none of the objects is in its original position.

For eg:- if the original arrangement is 1, 2, 3 then 2, 3, 1 is a derangement because no element remains in its original position.

The number of derangements of n objects is denoted by $!n$.

and the formula is: $!n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$

$$= !n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right).$$

Q105. Discuss the applications of counting principles in calculating probabilities and password security.

→ Counting principles are important in probability, password security, cryptography and many other areas.

1. Application in probability.

→ Counting principles help determine both the favorable and total outcomes.

Probability is calculated as $p(E) = \frac{\text{No. of favorable cases}}{\text{Total cases}}$.

For eg:- If a die is rolled and there are 6 possible cases.

$\{1, 2, 3, 4, 5, 6\}$

then, probability of getting even is $p(E) = \frac{3}{6} = \frac{1}{2}$.

2. Application in password security.

→ Counting principle determines the number of possible passwords. If a password contains 8 characters then no. of possible passwords is $26^8 = 208,827,064,576$.

A larger no. of possible passwords generally makes guessing or attacks more difficult and unpredictable.

Therefore, counting principles are used in probability and password security.

Unit-7 Algebraic structures.

Q131. Define a binary operation on a set. Give examples of operations that are and are not binary.

→ A binary operation on a non-empty set A is an operation that combines any two elements of A and produces an element that is also in A .

i.e. Binary operation $*$ on A is: $A \times A \rightarrow A$.

For eg:-

Binary operation.

Let set A contain element that are positive etc integers $\mathbb{Z}^+$.

i.e. $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Addition operation of set A is:-

$$\mathbb{Z} + \mathbb{Z} \rightarrow \mathbb{Z}$$

~~For~~ i.e. $3 + 5 = 8 \in \mathbb{Z}$

Therefore $+$ is an binary operation on A .

Operation that is not binary,

Subtraction of the set of natural number $\mathbb{N}$ is not necessarily binary.

For eg:- $2 - 5 = -3$

Since, $-3 \notin \mathbb{N}$, subtraction is not binary operation on $\mathbb{N}$.

Q132. What is an algebraic system (or structure)? Explain with an example.

→ An algebraic system or structure is a non-empty set together with one or more operation defined on that set.

It is generally written as: $(A, *)$ or $(A, +, \times) \dots$ where A is a set and the operations are defined on A .

For eg:-

The set of integers with addition is an algebraic structure: $(\mathbb{Z}, +)$.

Here, set = $\mathbb{Z}$ (integers).

operation = $+$ (Addition).

Let

$2+3=5 \in \mathbb{Z}$, Here, 2 and 3 are integers and after addition operation, result is also integers.

Q133. Define a semigroup. Give an example of a semigroup that is not monoid.

→ An semi-group is an algebraic structure consisting of a non-empty sets together with a binary operation $*$ that satisfies closure and associativity properties.

Thus, $(S, *)$ is a semigroup if:

1. Closure: For all $a, b \in S$, the result of the operation $a * b$ is also in S . ($a * b \in S$).

2. Associativity: For all $a, b, c \in S$, the equation $(a * b) * c = a * (b * c)$ is satisfied.

Monoid is a semigroup that also contains an identity element ($e \in S$: $a * e = e * a = a$ for all $a \in S$). Therefore, to find semi-group that is not monoid, it should lacks an identity element.

foreg:

Let set of positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ under addition operation ($+$).

closure: Adding any two positive integers gives another positive integer.

i.e. $a + b \in \mathbb{Z}^+$.

Associativity: Addition is naturally associative. i.e. $(a + b) + c = a + (b + c)$.

However, there is no identity element in $\mathbb{Z}^+$ because the additive identity is 0 and $0 \notin \mathbb{Z}^+$. Therefore, $(\mathbb{Z}^+, +)$ is a semigroup but not a monoid.

Q134. Define a monoid. What property distinguishes it from a semigroup?

→ Monoid is a semigroup that also contains an identity element.

Let set M with a binary operation $*$ is a monoid if it satisfies:

1. Closure: for all $a, b \in M$, $a * b \in M$.
2. Associative: for all $a, b, c \in M$, $(a * b) * c = a * (b * c)$.
3. Identity element: There exist an element $e \in M$ such that: $e * a = a * e = a$ for every $a \in M$.

Semi group	Monoid.
1. A semigroup only requires the properties of closure and associativity. 2. For eg: The set of natural numbers under addition operations without zero $(N, +)$ is a semigroup. As it is associative but it has no identity element.	1. Monoid is a semigroup that additionally contains a unique identity element. 2. For eg: The set of whole number under addition operation $(W, +)$ is a monoid because it includes identity element 0 (since $a + 0 = a$).

Q135. state the formal definition and four axioms of a group.

→ A group is a non-empty set G together with a binary operation $*$, written as $(G, *)$ satisfying four axioms.

Four group Axioms:

1. Closure: for all $a, b \in G$: $a * b \in G$.
2. Associativity: for all $a, b, c \in G$: $(a * b) * c = a * (b * c)$
3. Identity: There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.
4. Inverse: For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

For eg:

$(Z, +)$ is a group because every integer has an additive inverse.

Q126. What is an Abelian (commutative group)? prove that $(\mathbb{Z}, +)$ is an abelian group.

→ An abelian group is a group in which the binary operation is commutative.
i.e. for every $a, b \in G$ such that: $a * b = b * a$.

We need to verify all four group axioms and then commutativity.

1. Closure: Let $a, b \in \mathbb{Z}$. Since, the sum of two integers is an integer i.e. $a + b \in \mathbb{Z}$ for all $a, b \in \mathbb{Z}$.

2. Associativity: for all $a, b, c \in \mathbb{Z}$, $a + (b + c) = (a + b) + c$ which is associative for addition operation.

3. Identity: The identity element for addition is 0. for every $a \in \mathbb{Z}$, $a + e = e + a = a$, where identity element e is 0.

4. Inverse: for every $a \in \mathbb{Z}$, its additive inverse is $-a$. i.e. $-a \in \mathbb{Z}$. then $a + (-a) = 0$.

5. Commutativity: for all $a, b \in \mathbb{Z}$ such that $a + b = b + a$.

∴ All axioms of $(\mathbb{Z}, +)$ satisfies and it holds abelian group.

Q137. Define the order of a group and the order of an element within a group.

→ The order of a group G is the number of elements in the group. It is denoted by $|G|$.

For eg: if $G = \{0, 1, 2, 3, 4\}$ then $|G|$ is 5.

The order of an element $a \in G$ is the smallest positive integer n such that $a^n = e$ where, e is the identity element. It is denoted by $\text{ord}(a)$.

For eg: if $G = \{0, 1, 2, 3, 4, 5\}$ then $|G| = 6$.

For element 2, $\text{ord}(2) = 3$.

Q138. Define a subgroup. State the conditions required for a subset to be subgroup.

→ Let G be a group then a non-empty subset H of G is called a sub-group of G if H itself forms a group under the same operation as G .
It is denoted by $H \leq G$.

Conditions for a subgroup:

A non-empty subset $H \subseteq G$ is a subgroup if it holds:

1. Closure: for every $a, b \in H$: $a + b \in H$.
2. Associativity: It is automatically inherited from G .
for every $a, b, c \in H$: $a + (b + c) = (a + b) + c$.
3. Identity: The identity element of G must belong to H : $e \in H$.
4. Inverse: For every $a \in H$, $a^{-1} \in H$.

$\therefore$ Subgroup = Non empty set + closure + Identity + Inverse.

Q139. What is cyclic group? Give an example and identify its generator(s).

→ A group G is called cyclic group if all its elements can be generated by repeated applications of a single element and that element is called generators.

Let the group $G = (\mathbb{Z}_4, +_4)$ which consists of the set of remainders modulo 4 under the operation of addition modulo 4.

$\mathbb{Z}_4 = \{0, 1, 2, 3\}$, Operation $(+_4)$: add two numbers and take the remainder when divided by 4.

Identity element = 0.

To find the generators, we look for elements whose repeated addition generates the entire set $\{0, 1, 2, 3\}$.

1. Testing the element 1:

1. $1 \times 1 = 1$

2. $2 \times 1 = 1 + 1 = 2$

3. $3 \times 1 = 1 + 1 + 1 = 3$

4. $4 \times 1 = 1 + 1 + 1 + 1 = 4$ (In modulo 4, 4 becomes 0 since $4 \div 4$ leaves remainder of 0).

$\therefore \langle 1 \rangle = \{1, 2, 3, 0\}$ which formed original entire set.

2. Testing element 3:

1. $3 \times 1 = 3$

2. $3 \times 2 = 3 + 3 = 6 \pmod{4}$ gives $= 2$

3. $3 \times 3 = 3 + 3 + 3 = 9 \pmod{4}$ gives $= 1$

4. $4 \times 3 = 3 + 3 + 3 + 3 = 12 \pmod{4}$ gives $= 0$

$\therefore \langle 3 \rangle = \{3, 2, 1, 0\}$ which formed entire set.

$\therefore \langle 1 \rangle$ and $\langle 3 \rangle$ are generators.

Q140. Define rings and fields briefly. How do they extend the concept of a group?

→ A ring is a set R equipped with two binary operations usually called addition (+) and multiplication ($\cdot$). Such that $(R, +)$ is an abelian group and multiplication is associative:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

and multiplication distributes over addition:

$$a(b+c) = ab+ac$$

and

$$(a+b)c = ac+bc$$

Thus, a ring combines group structure under addition with multiplication.

A field is a commutative ring in which every non-zero element has a multiplication inverse.

for a field F :

$(F, +)$ is an abelian group, $(F - \{0\}, \times)$ is an abelian group and multiplication distributes over addition.

for every non-zero real number a , there exists:

$$a^{-1} = \frac{1}{a} \text{ such that } a \times \frac{1}{a} = 1.$$

Group has one operation whereas rings and fields have two operations. The progression from groups to rings and then to fields extends the foundational group concept by adding operations and structures:

1. From one operation to two: A group only defines one way to combine elements. Rings and fields introduce a second operation, allowing structures to model full arithmetic like standard number systems.
2. Connecting the operations: Rings introduce the distributive property. This acts as a bridge, dictating exactly how the two separate operations interact with each other.
3. Layering Group structures: A Ring contains exactly one group structure (under addition). Whereas field maximizes this concept by embedding two distinct group structure within single set as additive abelian group and multiplicative abelian group for all non-zero elements.

Q141. What is a Boolean Algebra? Define it using a set B , two binary operations and a unary operation.

→ A Boolean algebra is an algebraic structure that abstracts logical operations and set operations. It consists of non-empty set B $(B, +, \cdot, ')$ with two binary operations $+$ (logical OR / join) and $\cdot$ (logical AND / meet) and one unary operation $'$ (complement) and two distinct identity element 0 and 1 .

To qualify as Boolean Algebra the following axioms must hold for all elements $a, b, c \in B$.

1. Closure Laws.

i) closure under addition: $a + b \in B$.

ii) closure under multiplication: $a \cdot b \in B$.

2. Commutative Laws.

i) commutative under addition: $a + b = b + a$

ii) commutative under multiplication: $a \cdot b = b \cdot a$

3. Associative Laws.

i) $a + (b + c) = (a + b) + c$ (addition)

ii) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (multiplication)

4. Identity Laws:

i) Additive Identity (0): $a + 0 = a$

ii) Multiplicative Identity (1): $a \cdot 1 = a$

5. Distributive Laws:

i) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

ii) $a + (b \cdot c) = (a + b) \cdot (a + c)$

6. Complement Laws.

i) $a + a' = 1$

ii) $a \cdot a' = 0$

Q12. State the identity, complement, commutative and distributive laws of Boolean algebra.

→ Identity law states that a variable combined with an identity element retains its original value.

1. AND / Multiplication form: $A \cdot 1 = A$.

2. OR / Addition form: $A + 0 = A$.

Complement Law states that a variable combined with its exact opposite yields a constant result.

1. AND form: $A \cdot A' = 0$

2. OR form: $A + A' = 1$

Commutative Law states the order of the variables does not change the truth value of the expression.

1. AND form: $A \cdot B = B \cdot A$

2. OR form: $A + B = B + A$

4. Distributive law. states that operation can be distributed across operation inside parenthesis, similar to standard algebra.

1. AND over OR: $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

2. OR over AND: $A + (B \cdot C) = (A + B) \cdot (A + C)$.

Q143. Explain the principle of Duality in Boolean Algebra with an example.

→ The principle of Duality states that any true boolean algebraic expression remains true if operators and identity elements are systematically interchanged. To find the dual of an expression, make these exact swaps:

1. Change AND ($\cdot$) to OR ($+$) and vice-versa.

2. Change 0 to 1 and vice-versa.

For eg:-

Consider the valid boolean identity.

1. Distributive law: $A + (B \cdot C) = (A + B) \cdot (A + C)$.

Applying Duality principle, the expression becomes:-

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C).$$

Q144. What is Boolean function? How can it be represented using a truth table?

→ Boolean function is a mathematical function that maps input from a binary set to a binary output. Where inputs are n variables that can be '0' or '1', outputs is single value that is either '0' or '1' using logical operators (AND, OR, NOT).

For eg: $F(A, B) = A \cdot B + B'$.

Representation using a truth table:

1. First of all, total number of inputs is calculated using 2^n , where n is number of input variables.
2. placing each input variable using cycling binary combinations systematically.
3. Break the function into smaller logical steps.
4. Determine output based on the function's definition.

For eg:

$$\text{let } f(A, B) = A \cdot B + B'$$

$$\text{variables } (n) = 2$$

$$\text{then, total input rows} = 2^n = 2^2 = 4.$$

A	B	$A \cdot B$	B'	$A \cdot B + B'$
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

Fig: Truth table for $A \cdot B + B'$.

Q145. Explain sum of products (SOP) expansion (Disjunctive Normal Form).

→ A sum of products (SOP) expansion is a standard way of representing a boolean logical expression which consists of an OR operation of one or more AND operation. It is the product terms with logical AND variables where each variable appear exactly one which are connected by logical OR operations.

For eg:

$$f(A, B, C) = AB + C$$

expanding each term with missing variables.

$$\begin{aligned} F &= AB + C \\ &= AB(C + C') + \{C(A + A')\}(B + B') \\ &= ABC + ABC' + (AC + A'C)(B + B') \\ &= ABC + ABC' + ABC + A'B'C + AB'C + A'BC \\ &= ABC + A'B'C + AB'C + ABC' + A'BC \end{aligned}$$

Q146. Explain product of sums (POS) expansion (Conjunctive Normal Form).

→ A product of sums (POS) expansion is a method of representing a boolean logic function as AND operation of one or more OR terms.

It is the OR terms containing every variable of the function exactly one which is connected by AND operations.

Foreg: $f(A, B, C) = (A+B) \cdot (A'+C)$

expanding it with missing variable in each terms.

$$\begin{aligned} F &= (A+B) + (C \cdot C') \cdot (A'+C) + (C \cdot B \cdot B') \\ &= (A+B+C) \cdot (A+B+C') \cdot (A'+B'+C) \\ &\quad \text{[using distributive law]} \end{aligned}$$

$$\therefore F = (A+B+C) \cdot (A+B+C') \cdot (A'+B'+C) \cdot (A'+B'+C')$$

Q147. Write the Boolean expression $f(x, y, z) = x + yz$ in its complete sum of products form.

→ Given,

Boolean expression $f(x, y, z) = x + yz$.

Expand the each term with addition of missing terms.

$$\begin{aligned} F &= x + yz \\ &= x \cdot (y+y') + (yz) \cdot (x+x') \\ &= (xy + xy') \cdot (z+z') + xyz + x'yz \\ &= xyz + xy'z + x'yz + x'y'z + x'yz + x'y'z \end{aligned}$$

$$\therefore F = xyz + xy'z + x'yz + x'y'z + x'yz + x'y'z$$

Therefore, the complete sop form $= xyz + xy'z + x'yz + x'y'z + x'yz + x'y'z$.

Q148. Describe how logic gates (AND, OR, NOT) map to boolean algebra operations.

→ Logic gates directly represent basic Boolean algebra operations. The mapping of the three fundamental gates is:-

1. AND Gate → Boolean multiplication.

→ This gates acts as boolean multiplication and perform logical multiplication.

The output is true (1) only if all inputs are true.
It is represented by a dot (.) or by simply AB .

2. OR gate $\rightarrow$ Boolean addition.

$\rightarrow$ This gate acts as boolean addition which performs logical addition. The output of its gate is true (1) if at least one input is true (1). It is represented by a plus sign (+).

3. NOT gate $\rightarrow$ Boolean complement.

$\rightarrow$ This gate acts as boolean complement which performs logical inversion or complement. The output is the exact opposite of the input (0 becomes 1 and 1 becomes 0).
It is represented by bar over the variable ($\bar{A}$).

Q149. Explain the use of Karnaugh Maps (K-maps) for simplifying Boolean expression (up to 3 or 4 variables).

$\rightarrow$ Karnaugh maps (K-map) is a graphical tool used to simplify Boolean expressions without using complex algebraic theorems.

The uses of K-map :-

1. It eliminates the algebraic errors. It replaces the tedious boolean algebra theorems with visual pattern recognition.
2. It minimizes Hardware. It reduces the number of logic gates needed to build a physical digital circuit.
3. It standardizes output. It directly provides a minimized sum of products (SOP) or product of sums (POS) expression.

Q150. Simplify the boolean expression $F = AB + A\bar{B} + \bar{A}B$ algebraically.

$\rightarrow$ Given,

$$F = AB + A\bar{B} + \bar{A}B$$

$$= A(B + \bar{B}) + \bar{A}B$$

$$= A \cdot 1 + \bar{A} \cdot B \quad [\because \text{Complement law, } B + \bar{B} = 1]$$

$$= A + \bar{A}B$$

$$= (A + B)(A + \bar{A}) \quad [\because \text{Distributive law}]$$

$$F = 1 \cdot (A+B) \quad [\because \text{Complement law } A+\bar{A} = 1]$$

$$\therefore F = A+B$$

Therefore, the simplified boolean expression is $F = A+B$.

Q151. Discuss the importance of algebraic structures in cryptography and error-detecting codes.

→

Algebraic structures form the mathematical foundation of modern digital communication ensuring both security (cryptography) and reliability (error-detecting codes).

Importance in cryptography:-

i. Cryptography relies on problems that are easy to compute in one direction but computationally infeasible to reverse without a secret key. Algebraic structures provide the precise environment for these operations.

i) Groups: Cyclic groups, especially multiplicative groups of integers modulo a prime provide the basis for the Diffie-Hellman key exchange and digital signature algorithm.

ii) Fields: Finite fields are essential for symmetric-key cryptography. For eg: Advanced encryption standard (AES) performs all its byte-level substitutions and mix-column transformations using arithmetic over finite fields. This ensures highly non-linear transformations that resist linear and differential cryptanalysis.

iii) Elliptic curves: Elliptic curves over finite fields form an abelian group structure. It offers the same level of security as traditional RSA but with significantly smaller key sizes, reducing computational overhead and bandwidth for mobile devices and smart cards.

iv) Lattices: Advanced algebraic structures like Euclidean lattices form the foundation of post-quantum cryptography (PQC). Algorithms based on the shortest vector problem (SVP) in high-dimensional lattices are mathematically sound defenses against future quantum computer attacks.

Importances in Error-detecting codes.

When data is transmitted over noisy channels, it can be corrupted. Algebraic structures allow us to introduce redundant information systematically to detect and correct these errors.

1. vector spaces: Linear codes treat data blocks as vectors within a vector space over a finite field.

Encoding data is simplified to matrix multiplication

$(C = mG)$ where generator matrix G maps message vector to codewords. Detection relies on parity-check matrix H , where any non-zero result of CH indicates an error.

2. polynomial Rings: Cyclic codes are subclass of linear codes where any cyclic shift of a codeword is also a codeword. These codes are modeled using elements in a polynomial ring $P[x]/(x^n - 1)$.

Unit - 6: Graph Theory and Trees

Q106. Define directed, undirected, simple and multigraphs with examples.

→ Directed graph is a graph in which every edge has a specified direction from one vertex to another.



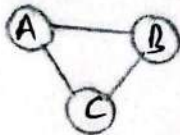
An undirected graph is a graph in which edges have no direction.

foreg:



A simple graph is an undirected graph that has no loops i.e. an edge from a vertex to itself and no multiple / parallel edges between the same pair of vertices.

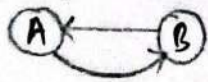
foreg:



→ edges AB, BC, CA with no loops or repeated edges.

A multigraph is a graph that can have multiple or parallel edges between the same pair of vertices.

foreg:

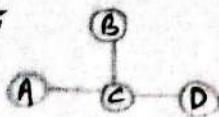


→ A and B are connected with parallel edges.

Q107. What is the degree of a vertex? state a handshaking theorem.

→ The degree of a vertex is the number of edges incident with that vertex. It is denoted by $\deg(v)$.

foreg:



for vertex C → $\deg(C) = 3$.

The handshaking theorem states that the sum of the degrees of all vertices in an undirected graph is twice the number of edges.

$$\text{i.e. } \sum_{v \in V} \deg(v) = 2|E|$$

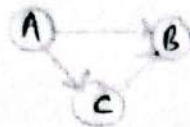
foreg: suppose a graph has 4 vertices with degrees: 2, 3, 2, 1
then, $\sum \deg(v) = 2 + 3 + 2 + 1 = 8$, $2|E| = 8 \therefore |E| = 4$

$\therefore$ The number of vertices having odd degree is always even.

Q108. Define in degree and outdegree in a directed graph. Show that the sum of in-degrees equals the sum of out-degrees.

→ The in-degree of a vertex is the number of directed edges entering that vertex. It is denoted by $\deg^-(v)$.
The out-degree of a vertex is the number of directed edges leaving that vertex. It is denoted by $\deg^+(v)$.

Foreg:



→ vertex A → $\deg^-(A) = 0, \deg^+(A) = 2$

→ vertex B → $\deg^-(B) = 2, \deg^+(B) = 0$

→ vertex C → $\deg^-(C) = 1, \deg^+(C) = 1$

Here, the sum of in-degree is:

$$\sum \deg^-(v) = 0 + 2 + 1 = 3$$

and, the sum of out-degree is:

$$\sum \deg^+(v) = 2 + 0 + 1 = 3$$

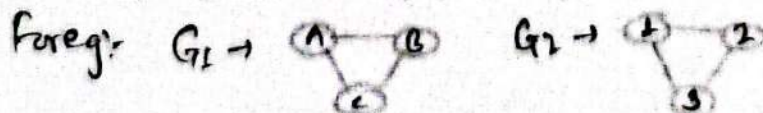
Hence, $\sum \deg^-(v) = \sum \deg^+(v)$. Therefore, the sum of in-degrees equals to the sum of out-degrees.

Q109. What are isomorphic graphs? state the necessary conditions for two isomorphic graphs.

→ Two graphs G_1 and G_2 are said to be isomorphic if there is one to one and onto correspondence between their vertices such that adjacency is preserved.

In simple words, two graphs are isomorphic if they have the same structure even if their vertices are named differently or drawn different.

It is written as: $G_1 \cong G_2$.



Necessary conditions to be isomorphic:

1. Same of vertices and edges for both graphs.
2. Same degree sequence for corresponding vertices.
3. Adjacency must be preserved: if two vertices are adjacent in one graph, their corresponding vertices must be adjacent.
4. Non-adjacency must also be preserved.

Q40. Define bipartite graphs and complete bipartite graphs ($K_{m,n}$).

→ A bipartite graph is a graph whose vertex set can be divided into two disjoint sets V_1 and V_2 such that every edge connects a vertex from V_1 to a vertex from V_2 .

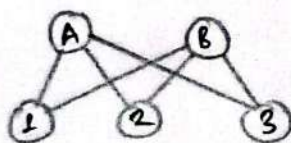
Foreg: Let, $V_1 = \{A, B\}$ and $V_2 = \{C, D\}$ Then



, All edges are between two sets.

Complete bipartite graph is a bipartite graph in which every vertex of one set is connected to every vertex of the other set. It is denoted by $K_{m,n}$ where, m is the number of vertices in first set and n is the number of vertices in second set.

Foreg: $K_{2,3}$. Let $V_1 = \{A, B\}$, $V_2 = \{1, 2, 3\}$

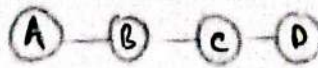


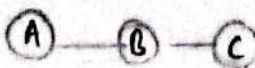
No. of edges = $A1, A2, A3, B1, B2, B3 = 6$.

No. of vertices = $m+n = 2+3 = 5$

Q41. What is sub-graph? Define proper graph and induced sub-graph.

→ Let $G = (V, E)$ be a graph and graph $H = (V_H, E_H)$ is called a sub graph of G if $V_H \subseteq V$, and $E_H \subseteq E$.

Foreg: If Graph $G \rightarrow$  then,

Graph $H \rightarrow$  is a subgraph of G .

A sub graph H of G is called a proper sub graph if H is not equal to G . At least one vertex or edge of G is missing from H . i.e. $H \subset G$

Foreg:

If G has vertices $\{A, B, C\}$ and edges $\{AB, BC, CA\}$ then a graph containing only AB and BC is a proper graph.

Induced graph is formed by selecting a subset of vertices from G and including all edges of G that connect those selected vertices.

Foreg:

Suppose $G : A - B - C$ and we select $V' = \{A, B\}$

The induced graph contains the edge AB , because it exists in G .

Q112. Explain paths, simple paths, circuits (cycles) and elementary cycles in a graph.

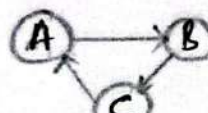
→ A path is a sequence of vertices in which consecutive vertices are connected by edges.

Foreg: $A - B - C - D$ is the path from A to D .

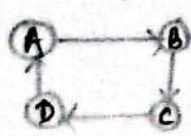
Simple path is a path in which no vertex is repeated.

Foreg: $A - B - C$ is a simple path because every vertex occur only once.

A circuit is a closed path in which the starting and ending vertices are same.

Foreg:  , A is the starting and ending vertex.

Elementary cycle is a cycle in which no vertex is repeated except the first and the last vertex.

Foreg: 

Q113. Differentiate between Strongly Connected, weakly Connected and unilaterally Connected digraph.

→ Strongly connected graph is a digraph if there is a directed path from every vertex to every other vertex.

For any two vertices u and $v : u \rightarrow v$ and $v \rightarrow u$ must be possible.

Foreg: $A - B - C - A$, Each vertex can reach to other vertex.

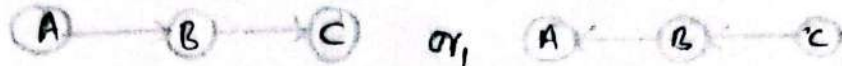
Weakly Connected graph is a digraph if it becomes connected when the direction of all edges are ignored.

Foreg:



unilaterally Connected graph is a digraph if for every pair of vertices u and v there is a directed path from u to v or from v to u . Both direction are not necessarily required.

Foreg:



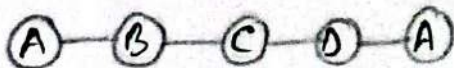
Q114. Define Euler paths and Euler circuits. State the condition for a graph to have an Euler circuit.

→ Euler path is a path that traverses every edges of a graph exactly once. vertices may be visited more than once, but every edge must be used exactly once.

Foreg:  can be Euler path if all edges are traversed exactly once.

Euler circuit is an Euler path that starts and ends at the same vertex.

Foreg:



Condition for Euler circuit.

A connected undirected graph has a Euler circuit if and only if every vertex has even degree.

$\deg(v)$ is even for every vertex v .

If there exists exactly two odd degree vertices then there exists Euler paths but not Euler circuits.

Q115. Define Hamiltonian paths and Hamiltonian circuits. How do they differ from Eulerian paths?

→ Hamiltonian path is a path that visits every vertex of a graph exactly once.

Hamiltonian Circuit is a Hamiltonian path that starts and ends at the same vertex forming a cycle.

Hamiltonian path	Eulerian path
1. It visits every vertex exactly once.	1. It visits every path/edge exactly once.
2. Some edges may be left unvisited.	3. Edges cannot be skipped or repeated.
3. Vertices cannot be repeated.	3. Vertices can be visited multiple times.

Q116. Explain the adjacency matrix and incidence matrix representations of a graph.

→

An adjacency matrix represents the connections between vertices (nodes) of a graph.

For a graph with n vertices, it is an $n \times n$ matrix where

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge between } v_i \text{ and } v_j \\ 0 & \text{if there is no edge between them} \end{cases}$$

For an undirected graph, the adjacency matrix is always symmetric ($A_{ij} = A_{ji}$).

The main diagonal elements (A_{ii}) are 0 unless the graph contains self-loops.

It requires $O(V^2)$ memory space, making it efficient for dense graphs but wasteful for sparse graphs.

An incidence matrix represents the relationship between vertices and edges of a graph.

For undirected graphs,

$$M_{ij} = \begin{cases} 1, & \text{if edge } j \text{ is incident on vertex } i \\ 0, & \text{if edge } j \text{ is not connected to vertex } i \end{cases}$$

It requires $O(V \times E)$ memory space where vertex (V) represents row and Edge (E) represent columns.

Q117. How is the number of paths of length k between two vertices calculated using an adjacency matrix?

→ Let A be the adjacency matrix of a graph with n vertices.

The entry $A_{ij} = 1$ if there is a direct edge from vertex i to vertex j .

The entry $A_{ij} = 0$ if there is no direct edge.

To find the number of paths of length k from vertex i to vertex j , compute the matrix power A^k by multiplying the adjacency matrix by itself k times.

$$A^k = \underbrace{A \times A \times A \times \dots \times A}_{k \text{ times}}.$$

The number of paths of length k from v_i to v_j is denoted by $(A^k)_{ij}$.

∴ Number of paths of length $k = (A^k)_{ij}$.

for eg:

$$\text{If } A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ then } A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

So, there is 1 walk of length 2 from v_1 to v_1 .

Q118. Define planar graphs. State Euler's formula for planar graphs ($v - e + f = 2$).

→ A planar graph is a graph that can be drawn in a single plane (such as a flat piece of paper) in a such way that no two edges cross or intersect each other except at their common vertices.

Euler's formula for planar graphs states that for any connected planar graph drawn in a plane without any edge crossings, the alternating sum of vertices, edges and face always equals a constant value of 2.

$$\text{is. } v - e + f = 2 \quad \text{where } \begin{aligned} v &= \text{number of vertices} \\ e &= \text{number of edges} \\ f &= \text{number of faces.} \end{aligned}$$

Q119. Explain the Breadth-First Search (BFS) algorithm and its application in finding shortest paths.

→ Breadth-First Search (BFS) is a graph traversal algorithm that visits vertices level by level, starting from a selected vertex. It uses a queue (FIFO).

Its algorithm:

1. Start from a source vertex.
2. Marked it as visited and insert it into a queue.
3. Remove a vertex from the queue.
4. Visit all its unvisited adjacent vertices and add them to the queue.
5. Repeat until the queue becomes empty.

Application in shortest paths.

In a unweighted graph, BFS finds the shortest path from a source vertex because it explores vertices according to their distance from the source.

Time Complexity: $O(V+E)$ where V = vertices and E = Edges.

Q120. Explain the Depth-First Search (DFS) algorithm and its use in finding connected components.

→ Depth-First Search (DFS) is a graph traversal algorithm that explores a graph by going as deep as possible before backtracking. It uses a stack or recursion.

Its algorithm:

1. Start from a selected vertex.
2. Mark it as visited.
3. Visit an unvisited adjacent vertex.
4. Continue deeper until no unvisited adjacent vertex remains.
5. Backtrack and continue with other unvisited vertices.

Application in finding connected Components:

It can find connected components by:

1. starting DFS from an unvisited vertex.
2. marking all vertices reachable from it.
3. starting another DFS from any remaining unvisited vertex.
4. Each DFS traversal represents one connected component.

Q12. Define a tree in graph theory. what are its fundamental properties?

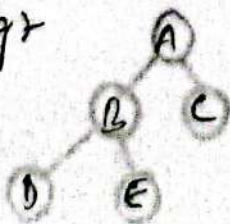
→ A tree is a connected undirected graph where there is a path between every pair of vertices and contains no cycles.

Fundamental properties:-

for any graph G with n vertices ($n \geq 1$), the following properties are need to be tree:

1. Unique path property: There is exactly one simple path between any two distinct vertices in a tree.
2. Edge count: A tree with n vertices always has exactly $n-1$ edges.
3. Minimally connected: A tree is connected but removing any single edge disconnects the graph.
4. Maximally Acyclic: A tree contains no cycles but adding any new edge between existing vertices creates exactly one cycle.
5. Bipartite Nature: Every tree is a bipartite graph meaning its vertices can be divided into two independent sets such that no two graph vertices within the same sets are adjacent.

For eg:



vertices = 5 , edges = 4

Q122. Define root, internal vertices, leaf nodes and the height (depth) of a rooted tree.

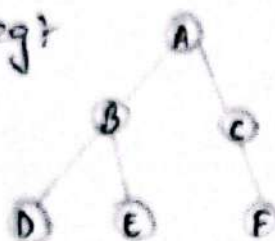
→ A root is a unique, designated vertex in a tree from which all hierarchical relationships originate.

An Internal vertex is any vertex of a rooted tree that has at least one child.

A Leaf node is a vertex that has no children and represents the end of a branch in the hierarchical structure.

The height of a rooted tree is the maximum length of a path from the root to any leaf node.

foreg:



→ The topmost node is a root is A.

→ Internal vertices are A, B, C.

→ Leaf nodes → D, E, F

→ Height of rooted tree:

$$A \rightarrow B \rightarrow D = 2$$

$$A \rightarrow B \rightarrow E = 2$$

$$A \rightarrow C \rightarrow F = 2$$

$$\therefore \text{Height} = 2.$$

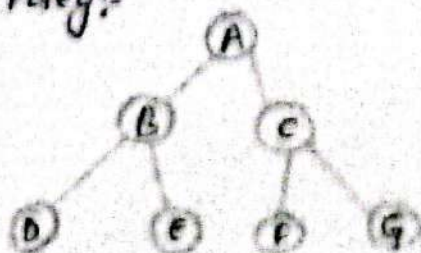
Q123. What is an m-ary tree? Define a full binary tree and complete binary tree.

→ An m-ary tree is a rooted tree in which each internal vertex has at most m children.

foreg: In 3-ary tree, each node can have at most 3 children.

A full binary tree is a binary tree in which every internal node has exactly two children.

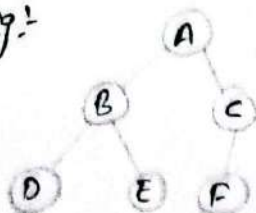
foreg:-



Every internal node (A, B, C) has exactly two children.

A complete binary tree is a binary tree in which every level except possibly the last is completely filled and nodes in the last level are filled from left to right.

For eg:-



This is complete because the last level is filled from left to right.

Q124. If a full m -ary tree has P internal vertices, prove that it has $n = m \cdot i + 1$ total vertices.

→ Let n be the total number of vertices in the tree, P be the number of internal vertices and m be the number of children of each internal vertex.

In a full m -ary tree, every internal vertex has exactly m children.

Therefore, total edges, $e = m \cdot i$.

and every tree satisfies, $e = n - 1$ for every n vertices.

$$\therefore e = n - 1$$

Hence,

$$n - 1 = m \cdot i$$

$$\text{or, } n = m \cdot i + 1$$

Therefore, $n = m \cdot i + 1$. proved.

Q125. Explain preorder tree traversal. Write the sequence for a sample binary tree.

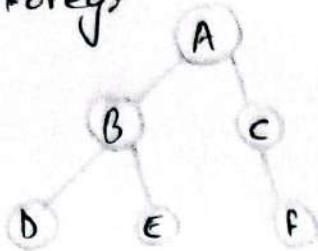
→ preorder tree traversal is a depth-first tree traversal method where each node is visited before its subtrees.

The traversal follows: $\text{Root} \rightarrow \text{Left subtree} \rightarrow \text{Right subtree}$.

To perform a preorder traversal, we need to follow three steps recursively for each node:

1. visit the root node : process or print the data of the current node.
2. Traverse the left subtree : move to the left child and repeat the process.
3. Traverse the right subtree : move to the right child and repeat the process.

Foreg:



Traversal process:

1. visit Root A.
 2. Go left to Node B, then D and then E.
 3. Go right to Node C and then visit root F.
- ∴ final preorder Sequence : $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F$.

Q12. Explain Inorder tree traversal. write the sequence for a sample binary tree.

→ It is a type of Depth-first search (DFS). When applied to a Binary search Tree (BST), inorder traversal visits the nodes in ascending sorted order. The traversal follows:

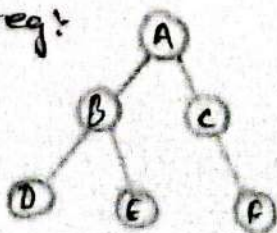
Left subtree → Root → Right subtree.

Algorithm:-

For any given node the algorithm performs three steps recursively.

1. visit the left subtree first.
2. visit the current root node.
3. visit the right subtree last.

Foreg:



Traversal steps:

1. Go to left subtree of A are D, B, E
2. Back to visit A.
3. Go to right subtree of A are C, F.

∴ In order sequence : $D \rightarrow B \rightarrow E \rightarrow A \rightarrow C \rightarrow F$.

Q127. Explain Postorder tree traversal. Write the sequence for a sample binary tree.

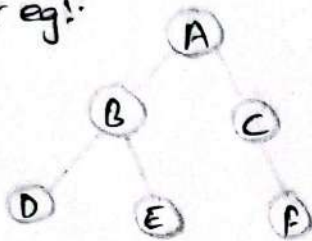
→ It is a depth-first algorithm used to visit every node in a binary tree. In this, the root node is visited last after visiting its entire left and right subtrees.

The traversal follows Left subtree → Right subtree → Root.

Algorithm: For every given node, the algorithm works recursively:

1. Traverse the left subtree recursively in postorder.
2. Traverse the right subtree recursively in postorder.
3. Visit the root node to read or process its value.

For eg:



Traversal steps:

1. Go to its left subtree of A → D, E, B
2. Go to its right subtree → F, C
3. Visit the root A.

postorder sequences: D → E → B → F → C → A.

Q128. What is spanning tree? Explain how to find it using BFS or DFS.

→ A spanning tree is a connected, acyclic subgraph of an undirected, connected graph $G(V, E)$ that includes all vertices V of the graph with the minimum possible number of edges $(V-1)$.

Finding a spanning Tree using BFS (Breadth-first search).

1. Select a starting vertex.
2. Visit all adjacent unvisited vertices.
3. Add the traversed edges to the spanning tree.
4. Continue level by level until all vertices are visited.

Finding a spanning tree using DFC. (Depth-First search)

1. start from any vertex.
2. move as deep as possible along unvisited vertices.
3. Add traversed edges to the spanning trees.
4. backtrack when necessary.

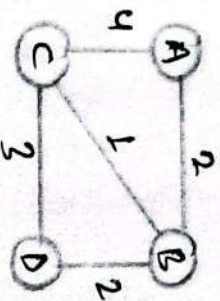
Q119. Briefly describe Minimum spanning Trees (MST) and name one algorithm used to find it. (eg. Prim or Kruskal's).

→ A minimum spanning tree of a weighted connected graph is a spanning tree whose total edge weight is minimum among all possible spanning trees.

Its properties:

1. It contains all vertices.
2. It contains exactly $n-1$ edges.
3. It has no cycles.
4. It includes minimum total cost (weight).

For eg



MST selects edges with minimum total weight.

Algorithms used.

1. Prim's Algorithm.

→ i) starts from any vertex.

ii) Repeatedly adds the minimum-weight edge connecting a new vertex.

2. Kruskal's Algorithm.

→ i) sort edges by weight.

ii) Add the smallest edge that does not form a cycle.

Q120. How are trees utilized in a computer science? Discuss Binary search trees (BST) or Abstract Syntax Trees?

→ Applications of trees in computer science
Trees are widely used in:-

1. File and directory systems.
2. Database indexing (B-Trees, B+ trees).
3. Expression Evaluation.
4. Routing algorithms.
5. Compiler design.
6. Searching and sorting operations.

1. Binary Search Trees (BST)

→ It is a node-based binary tree data structure that enforces a strict ordering rule to optimize data retrieval.

For any given node, all values in its left subtree must be less than the node's value and all values in its right subtree must be greater than the node's value.

They are primarily used for efficient searching, insertion and deletion operations. In a balanced BST, search operations take $O(\log n)$ time complexity.

2. Abstract Syntax Trees (AST)

→ It is a hierarchical representation of the abstract syntactic structure of a source code written in a programming language.

Each node in the tree denotes a construct occurring in a source code (such as operator, a loop, or a variable definition).

Compilers and interpreters use AST to break down source code into structured format that machine can analyze, optimize and translate.

For eg:-

Expression: $(a+b) \times c$

AST is

